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Project Proposal

AI-Based Optimisation of the 200 MLD Pirana Sewage Treatment Plant
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1. EXECUTIVE OVERVIEW

The Pirana 200 MLD Sewage Treatment Plant is one of Ahmedabad's most critical municipal assets, responsible for treating large volumes of city wastewater every day before it is released back into the environment. The facility has recently been upgraded with Integrated **Fixed-Film Activated Sludge (IFAS) technology** and advanced biofilm media cages to improve biological treatment and meet the **National Green Tribunal (NGT)** 2019 discharge standards. These physical upgrades have created a strong treatment foundation, but the plant is still operated largely through manual decisions and fixed set-points, which prevents the full potential of the new system from being realised.

Today, key processes such as aeration, sludge recycling, chemical dosing, and disinfection are mostly controlled based on operator experience and conservative safety margins. This approach leads to several challenges: higher-than-necessary energy consumption, inconsistent effluent quality, difficulty in maintaining nitrogen and phosphorus limits, and limited early warning when equipment or biological processes begin to deteriorate. In particular, blower, pumps and motor systems often run at or near full capacity, and chemical dosing is typically fixed rather than adjusted based on real-time conditions. As a result, the plant achieves compliance most of the time, but not with the consistency, efficiency, or predictability expected from a modern, upgraded facility.

Logicrest Technologies proposes an AI-driven optimisation and automation layer that sits on top of the existing infrastructure and recent IFAS upgrade.The system connects to the sensors and control panels, collects data continuously, and uses specialised AI and machine learning models to calculate the most efficient and compliant way to run the plant at each moment. A custom-built software platform developed by Logicrest will provide a single unified “control cockpit” for the plant, while the SCADA system acts as a reliable mediator between field instruments, control logic, and operator commands.

The solution will use specialised AI models to optimise all major treatment stages, including aeration, sludge recycling, chemical dosing, digestion, and disinfection. These models will forecast treatment loads, adjust set-points such as dissolved oxygen and sludge wasting rate, detect early signs of equipment wear or process instability, and provide clear, explainable recommendations to operators. Over time, once operators gain confidence, the system can move from advisory mode to semi-autonomous and then fully autonomous operation, always preserving manual override capability for safety.

For investors and senior decision-makers, the value proposition is straightforward: this project converts an already capital-intensive, upgraded IFAS plant into a smart, self-optimising utility asset. The AI layer is designed to enhance regulatory compliance, reduce operational risk, stabilise performance in any condition, and unlock significant savings without any compromise on treatment quality. In summary, the proposal is not an IT experiment; it is an operational transformation that uses AI and modern software to protect prior infrastructure investments, strengthen environmental performance, and create a robust, future-ready wastewater treatment benchmark for Ahmedabad.

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2. PROJECT DEFINITIONS

Project Name

AI-Driven Optimisation System for Pirana 200 MLD STP - Complete Facility Integration

***Clarification**:* “Optimisation” here means continuously finding the most efficient and reliable way to operate the plant using data and algorithms, rather than changing the physical treatment units.

Project Scope

Deployment of integrated artificial intelligence and automation technologies across:

- **Primary clarification process** - Intelligent influent analysis and predictive load forecasting through continuous monitoring of BOD, COD, pH, TSS, temperature, and flow patterns, enabling early detection of shock loads and proactive adjustment of downstream biological processes.

- **Activated Sludge Aeration System (with IFAS Media)** - Biofilm-aware dissolved oxygen control and biofilm health monitoring, where the AI continuously adjusts aeration based on real-time load, temperature, and biofilm behaviour to avoid over- or under-aeration.
- **Secondary clarification and sludge management** - Mixed Liquor Suspended Solids (MLSS) control and integrated sludge handling optimisation, so that return sludge (RAS) and waste sludge (WAS) flows are adjusted intelligently to maintain the correct biomass concentration and settling characteristics.
- **Chemical dosing optimisation** - Adaptive coagulant and polymer dosing that responds to actual sludge properties and effluent targets instead of fixed, conservative doses, thereby reducing wastage and improving sludge dewatering performance.
- **Disinfection control** - Chlorine residual optimisation to achieve target microbial standards with minimal chemical use by linking chlorine dose to real-time effluent quality and contact time.
- **Biofilm Media Health Monitoring** - Real-time assessment of installed IFAS media performance and stress detection using indirect indicators such as oxygen transfer efficiency, ammonia removal performance, and sludge characteristics.
- **Advanced sludge treatment** - Digester optimisation and bulking prevention, where AI models balance sludge age, loading, and mixing to stabilise digestion, support biogas generation potential, and prevent sludge bulking events.
- **Real-time monitoring and predictive analytics** - Comprehensive process intelligence across the entire plant, enabling forecasting of loads, early detection of anomalies, and clear decision support for operators and management.
- **SCADA software integration in custom software**- A purpose-built Logicrest software platform will act as the primary operations and analytics environment, while the SCADA system functions as a secure mediator between field instruments, controllers, and the Logicrest software platform. All control commands will be routed through SCADA for safety, standardisation, and compliance with existing plant architecture.

System Components

Hardware Infrastructure:

- Edge computing servers (on-site deployment) with industrial-grade reliability for running the Logicrest AI platform and handling real-time data processing
- HART protocol multiplexing (Highway Addressable Remote Transducer—a digital communication protocol that allows multiple sensors to transmit measurement data over a single cable pair) for data acquisition from installed sensor network

- Network infrastructure and cybersecurity equipment (firewalls, secure gateways) to ensure secure communication between SCADA, Logicrest software platform, and field devices
- Integration with existing Endress+Hauser sensor network (already installed: 30+ critical process instruments across inlet, aeration, secondary clarifier, and sludge streams)
- Additional monitoring hardware for IFAS biofilm assessment, digester optimisation, and predictive load forecasting at primary clarification
- SCADA integration infrastructure serving as the secure mediator layer between field instruments, PLCs/controllers, and the Logicrest software platform

Software Platform:

- **Custom Logicrest Software Platform (Primary operations and analytics environment):**
  - Real-time data visualisation and comprehensive process monitoring dashboard
  - Integrated control logic for DO, MLSS, aeration, RAS, WAS, chemical dosing etc., and predictive load management
  - Alarm management and intelligent operator advisory system
  - Historical data logging and advanced performance analytics
  - User interface with role-based access (Operator, Supervisor, Engineer etc.)
  - Predictive forecasting engine
- **AI/ML models for process Optimisation:**
  - **Core modules:** Baseline activated sludge process optimisation (DO control, MLSS management, energy optimisation, chemical dosing)
  - **IFAS-specific modules:** Nitrification/denitrification prediction, biofilm health monitoring, biofilm stress detection, temperature-driven seasonal adjustments
  - **Advanced optimisation modules:** Inter-dependent parameter control (MLSS-SRT-F/M coupling), shock load response, nutrient balance monitoring
  - **Sludge management modules:** Bulking prevention, digester optimisation, biogas production forecasting
- **SCADA System Integration (Secure mediator role):**
  - Acts as the standardised communication bridge between field instruments/PLCs and the Logicrest software platform
  - Routes all control commands from Logicrest AI through SCADA for safety, audit compliance, and alignment with existing plant control architecture

    - Provides real-time sensor data streams to Logicrest software platform via secure API interfaces
    - Maintains independent safety interlocks and manual override capability

- Real-time monitoring dashboard with biofilm health indicators
- Predictive maintenance and anomaly detection systems
- Operator interface with explainable AI recommendations (SHAP-based)
- Cloud-based data storage and analytics (with on-premises backup for data sovereignty)
- Biofilm-aware process control algorithms

Integration Services:

– Development and deployment of complete custom Logicrest software platform as primary control and analytics system

– Configuration of SCADA as secure mediator between field devices and Logicrest software platform (routing control commands, sensor data streaming)

– Integration with existing Endress+Hauser sensor data streams via SCADA communication protocols

– Development of custom APIs and data bridges connecting SCADA ↔ Logicrest software platform ↔ cloud analytics

– Operator training and change management (Logicrest software platform interface, IFAS-specific features, predictive load forecasting, AI recommendations interpretation etc.)

– Performance validation and acceptance testing across all integrated systems (SCADA mediator functionality, Logicrest AI control accuracy, sensor integration)

– 12-month warranty and support period with quarterly optimisation reviews and model retraining

Project Duration

Total Implementation Timeline: 15 Months

- Phase 1 - Design & Foundation: 3 months
- Phase 2 - Hardware Installation & SCADA Integrated Software Development: 3 months
- Phase 3 - Core Model Deployment & Testing: 3 months
- Phase 4 - Advanced Modules & Optimisation: 3 months
- Phase 5 - Full Autonomy & Stabilisation: 3 months

Stakeholders

- **Project Beneficiary:** Ahmedabad Municipal Corporation (AMC)
- **Operator:** DNP Infrastructures (O&M improvement)
- **Technology Provider:** Logicrest Technologies Pvt Ltd
- **Design Review Partner:** IITRAM (Indian Institute of Technology, Rajasthan)
- **End Users:** Ahmedabad residents and stakeholders

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3. CURRENT PROBLEM STATEMENT

1. Operational Inefficiencies & Energy Consumption

1.1 Fixed - Speed Blower Operation

Current Status: Aeration blowers operate at or near 100% capacity continuously, regardless of actual treatment demand, time of day, or seasonal variations.

Issue: No intelligent real-time mechanism exists to adjust blower output to match biological oxygen demand. The system over-aerates during low-load periods (nights, weekends, low-flow seasons) as if operating at peak load.

Impact:

- 30–40% of blower energy is wasted due to mismatch between air supply and oxygen demand
- Aeration accounts for 30–40% of total facility energy; this inefficiency translates to substantial recurring avoidable costs

Solution:

AI-driven adaptive aeration control (DO Set-point Optimisation Model, Aeration Demand Forecasting Model in Section 5 AI/ML Models) will continuously adjust blower speed based on real-time load, reducing energy consumption by 15–30% while maintaining treatment quality.

1.2 Slow and Manual Dissolved Oxygen (DO) Control

Current Status: Dissolved oxygen (DO) levels controlled manually through intermittent lab tests and operator judgment. Conservative DO set-points maintained at 3–4 mg/L to avoid under-aeration risk.

Issue:

- Manual adjustments are reactive and delayed
- DO fluctuates significantly due to slow operator response
- System experiences alternating periods of over-aeration (energy waste) and under-aeration (treatment risk)
- No biofilm-aware control for IFAS media

Impact:

- Inconsistent biological performance with nitrification efficiency at 70–80%
- Energy waste during over-aeration periods
- Risk of ammonia breakthrough during under-aeration episodes
- IFAS system underutilised: biofilm potential not fully leveraged despite capital investment

Solution:

Real-time AI control (Models 1, 9, 10 as further mentioned in 5th AI/ML Models section) will stabilise DO within ±0.5 mg/L, optimise for biofilm + suspended biomass simultaneously, and respond to load changes within 5–15 minutes instead of 30–60 minutes.

2. Chemical Overdosing & Unoptimised Chemical Control

2.1 Fixed Polymer and Coagulant Dosing

Current Status: Polymer and $FeCl_3$ (coagulant) dosed using fixed set-points based on historical assumptions and conservative safety margins. Doses rarely adjusted for real-time sludge characteristics or influent variability.

Issue:

- Safety margin approach leads to persistent overdosing
- No active control to exploit coagulant-polymer synergy
- Fixed doses do not respond to daily changes in solids loading, sludge characteristics, or influent BOD/turbidity

Impact:

- 15–25% polymer waste (unnecessary chemical expenditure)
- 15–20% $FeCl_3$ waste (overdosing with no treatment benefit)

- Higher sludge handling volumes due to excess chemical addition

Solution:

AI-driven chemical optimisation (Chemical Dosing Optimisation Model in Section 5 AI/ML Models) will monitor real-time sludge characteristics and adjust polymer/coagulant pump speeds dynamically, eliminating safety margin overdosing while maintaining target cake dryness.

2.2 Chlorine Dosing Without Residual-Based Control

Current Status: Disinfection managed through fixed chlorine dose (typically 1.8–2.2 mg/L) to ensure regulatory compliance under worst-case conditions.

Issue:

- Fixed-dose strategy does not respond to variations in effluent quality (BOD, TSS, turbidity)
- No systematic tracking of CT (concentration × time) relationship
- Plant maintains higher residual than needed to achieve disinfection targets

Impact:

- 15–25% chlorine waste compared to optimised residual-based dosing
- Higher residual chlorine in effluent than environmentally necessary

Solution:

Chlorine residual optimisation (Model 20 in Section 5 AI/ML Models) will link chlorine dose to real-time effluent quality and contact time, reducing chemical usage by 15–25% while maintaining <100 CFU/100mL compliance.

3. Compliance & Regulatory Risk (NGT 2019 Standards)

3.1 Inconsistent Effluent Quality

Current Status: Plant meets NGT 2019 standards most of the time but not consistently year-round. Observed effluent values:

- BOD: 8–15 mg/L (target: <10 mg/L)
- Ammoniacal Nitrogen: 3–8 mg/L (target: <5 mg/L)

- Total Nitrogen: 12–18 mg/L (target: <10 mg/L)
- TSS: 8–12 mg/L (target: <10mg/L)

Issue:

System vulnerable to:

- Shock loads (sudden organic or nitrogen spikes)
- Seasonal temperature changes
- Operational disturbances (equipment downtime, manual adjustment delays)
- Metropolitan/mega city standards are more stringent than Tier 1 cities (TN <10 mg/L vs. <15 mg/L), requiring tighter process control.

Impact:

- Current compliance rate: 85–92% (approximately 1–2 violations per year)
- Risk of NGT penalties and regulatory action
- Environmental non-compliance impacts public perception and stakeholder confidence
- Winter months particularly vulnerable: Ammoniacal Nitrogen frequently exceeds 5 mg/L and Total Nitrogen frequently exceeds 10 mg/L

Solution:
Predictive control (Models 5, 9, 10, 14 in Section 5 AI/ML Models) will forecast effluent quality 1–2 days ahead, enabling proactive adjustments before violations occur. Target: 95% Year 1 compliance with metropolitan/mega city standards (BOD <10 mg/L, NH_3-N <5 mg/L, TN <10 mg/L, TSS <10 mg/L); 99%+ Year 2+.

3.2 Winter Vulnerability and Nitrification Limitations

Current Status: Ahmedabad winter temperatures significantly slow nitrifying bacteria growth and activity.

Issue:

- Without compensating adjustments in SRT, MLSS, and DO, nitrification efficiency drops 40–50% in winter
- Current semi-manual approach cannot predict or proactively counteract seasonal decline

- IFAS system installed for better cold-weather nitrification but lacks AI-driven seasonal control to leverage it

Impact:

- Ammonia and Total Nitrogen levels drift upward Dec–Feb (often exceeding 5 mg/L NH_3-N and 10 mg/L TN respectively)
- Compliance margin disappears during precisely the period requiring most precise control
- Predictable winter NGT violations without intelligent seasonal adaptation

Solution: Temperature-driven seasonal SRT adjustment (Model 14 in Section 5 AI/ML Models) will automatically increase sludge age by 50–100% during winter, maintaining 75–80% nitrification efficiency. IFAS biofilm provides cold-stable nitrifier population.

4. Sludge Management Challenges

4.1 Manual and Reactive SRT Control

Current Status: Solids Retention Time (SRT) managed manually via WAS flow adjustments based on operator judgment and intermittent lab measurements. No continuous, data-driven SRT optimisation.

Issue:

- SRT too low: Insufficient nitrifier population → ammonia breakthrough
- SRT too high: Filamentous growth risk, increased oxygen demand, energy waste from excessive biomass

Impact:

- Periods of ammonia non-compliance when SRT inadequate
- Periods of increased energy consumption when SRT excessive
- Inconsistent biological population quality and treatment stability

Solution:
Coupled MLSS-SRT-F/M optimisation (Model 13 in Section 5 AI/ML Models) will continuously calculate optimal SRT based on temperature, load, and nitrification requirements, automatically adjusting WAS flow to maintain ideal sludge age.

4.2 Sludge Bulking & Lack of Early Warning

Current Status: Sludge bulking detected only after clarifier performance visibly degrades (high sludge blanket, effluent TSS increase).

Issue:

- No predictive mechanism analyses trends in F/M ratio, DO, temperature, or SRT
- Filamentous microorganisms proliferate unchecked until crisis occurs
- High Sludge Volume Index (SVI) causes poor settling and solids carryover

Impact:

- Corrective actions taken too late (after clarifier already compromised)
- Recovery is slow and operationally disruptive (1–2 weeks typical)
- Risk of effluent quality violations during bulking events
- Lost treatment capacity during recovery period

Solution: Sludge bulking early warning (Model 15 in Section 5 AI/ML Models) will detect filament growth risk 3–7 days in advance by monitoring F/M, DO, temperature, and SRT trends. Enables preventive action before SVI exceeds 120 mL/g.

4.3 Digester Performance & Biogas Utilisation

Current Status: Anaerobic digester operates with fixed feed rates and mixing intensity. No predictive monitoring for digester health or optimisation for biogas production.

Issue:

- Feed rates not optimised based on solids loading, volatile solids content, or digester health
- Fixed mixing intensity regardless of process needs
- No early warning for digester upset (pH drops, biogas production decline, temperature deviations)

Impact:

- Biogas production 15–20% below theoretical potential (lost energy recovery opportunity)
- Occasional digester instabilities requiring 5–10 day recovery periods

- Reduced sludge stabilisation efficiency
- Lost opportunity to offset facility energy costs through biogas utilisation

Solution:
Digester optimisation (Models 17, 18 in Section 5 AI/ML Models) will dynamically adjust feed rate and mixing intensity based on biogas production, temperature, and volatile solids destruction. Predicts upsets and automatically reduces feed to enable recovery.

4.4 Chemical Phosphorus Levels > 2 mg/L in Biosolids

Current Status: Due to coagulant ($FeCl_3$) usage, residual phosphorus in biosolids consistently measured at >2 mg/L.

Issue:

- High chemical phosphorus from iron/aluminium accumulation in sludge
- No closed-loop control linking coagulant dosing to long-term biosolids phosphorus levels
- Coagulant dosing not optimised, leading to persistent chemical accumulation

Impact:

- Biosolids quality limited to Class B (restricted agricultural reuse)
- Cannot upgrade to Class A biosolids (unrestricted use) due to phosphorus barrier
- Long-term accumulation increases inert, non-biodegradable solids fraction
- Reduces effective treatment capacity and increases disposal costs
- Missed revenue opportunity: Class A biosolids command premium pricing

Solution:
Enhanced chemical optimisation (Model 4 in Section 5 AI/ML Models) will reduce $FeCl_3$ usage by 15–20% through improved polymer synergy, gradually lowering biosolids phosphorus from >2 mg/L toward ≤1.0 mg/L over 6–12 months, opening pathway to Class A classification.

4.5 SCL (Sludge Concentration Level) Measurement Limitations

Current Status: Sludge Concentration Level (SCL) inferred from turbidity, periodic lab TSS tests, or occasional sensor readings. Not continuously measured with high accuracy.

Issue:

- Indirect or infrequent measurement influenced by particle size, sludge characteristics, sensor fouling
- Leads to uncertainty in actual solids concentration (±10–20% error common)
- RAS and WAS flow decisions based on approximate SCL values

Impact:

- SCL overestimation: Insufficient RAS adjustment → excessive MLSS → increased oxygen demand and energy waste
- SCL underestimation: Low MLSS → reduced treatment capacity → poor effluent quality
- Difficult to maintain optimal biomass concentration in aeration tanks
- Increased risk of clarifier overloading and solids carryover
- Inconsistent sludge withdrawal complicates SRT management

Solution: Virtual SCL modelling (integrated in Model 13) will combine available measurements with MLSS data, flow patterns, and historical trends to infer reliable solids concentration. Enables smarter RAS/WAS control even with imperfect direct measurement.

5. Monitoring, Visibility & Decision-Making Gaps

Current Status: Fragmented data from sensors and laboratory tests without unified analysis or predictive capability.

Issue:

- No single integrated platform translates complex process data into actionable insights
- High dependence on individual operator experience → variability in decisions and outcomes
- Operators react after problems appear rather than preventing them

Impact:

- Limited ability to guarantee consistent NGT 2019 compliance
- Cannot minimise energy and chemical costs systematically
- No proactive equipment protection (failures discovered after occurrence)
- Cannot progress toward higher biosolids quality (Class A) without unified strategy

Solution: Comprehensive AI-driven automation system (specialised models + custom SCADA integrated logicrest platform) will unify all data streams, provide predictive insights 1–4 weeks ahead, deliver real-time operator recommendations, and enable 24/7 autonomous optimisation with explainable logic.

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4. PROJECT AIM & SOLUTION APPROACH

Core Objectives

Primary Aim: To implement an integrated, AI-driven automation and optimisation system that transforms the Pirana 200 MLD STP from a semi-manual, reactive operation into a stable, self-optimising, and compliance-assured facility. The system will harness the existing IFAS upgrade, installed sensors, and process equipment to deliver:

- **Operational efficiency through real-time adaptive control**
  All major treatment units (aeration, clarification, sludge handling, digestion, and disinfection) will be controlled using continuously updated set-points, not fixed manual values. The AI system will respond to changing flow, load, and temperature conditions within minutes, rather than the current 30–60 minute manual reaction time.
- **Consistent regulatory compliance with NGT 2019 standards**
  The system will maintain effluent BOD, TSS, and Total Nitrogen within regulatory limits(BOD <10 mg/L, TSS <10 mg/L, TN <10 mg/L), even during winter and shock load events, by predicting future performance and adjusting process conditions proactively instead of reacting after violations occur.
- **Maximum utilisation of IFAS biofilm-based treatment enhancement**
  The IFAS media already installed at the plant is capable of advanced nitrification and simultaneous nitrification–denitrification (SND). The AI will explicitly model biofilm behaviour, oxygen transfer, and temperature effects so that both the suspended sludge and biofilm phases are operated at their optimal conditions.
- **Energy optimisation and chemical cost reduction**
  The system will target a holistic reduction in total plant energy use, not just blower power. This includes aeration, pumps, mixers, and digester systems, alongside optimisation of polymer, coagulant, and chlorine dosing to eliminate safety-margin overdosing.
- **Predictive asset protection and reduced equipment failures**
  All critical rotating equipment (blowers, pumps, mixers, centrifuges) will be monitored for

early signs of failure using vibration, power, and operating trend analysis, allowing maintenance to be planned weeks in advance instead of after breakdown.

- **Predictive performance and early intervention capability**
  The AI models will forecast key performance indicators (effluent BOD, TN, ammonia, SVI, biogas production, energy use) hours to days ahead, giving operators a clear warning window to intervene before problems translate into non-compliance or process upset.
- **Transition from reactive manual operation to strategic autonomous control**
  Routine control actions (adjusting DO, RAS/WAS, chemical doses, digester feed, chlorine) will be handled autonomously by the AI, while operators focus on supervision, strategic decision-making, process improvements, and long-term planning.
- **Biosolids quality upgrade from Class B to Class A**
  Currently, the plant produces Class B biosolids, which have restricted reuse and lower value. By optimising sludge age, digestion, and chemical dosing (especially phosphorus-related coagulants), the project aims to move towards Class A biosolids with higher stabilisation, lower chemical residues, and better suitability for beneficial reuse.

Key Performance Objectives

1. Energy Efficiency

Objective: Reduce overall plant energy consumption by an estimated 15–30% through AI-driven aeration optimisation, intelligent blower staging, adaptive pump scheduling, IFAS mixing optimisation, and load-responsive operation across treatment units.

How This Will Be Achieved:

- Aeration optimisation: DO set-points adjusted every few minutes based on real-time load and predicted nitrification needs, preventing chronic over-aeration.
  - Intelligent blower staging: Instead of running multiple blowers at fixed high speeds, the AI will select the optimal combination of blowers and speeds to meet air demand with minimum power.
  - Adaptive pump scheduling: Influent, RAS, WAS, sludge transfer, and digester feed pumps will be scheduled and modulated to avoid unnecessary run-time, peak demand spikes, and inefficient partial-load operation.

- IFAS mixing optimisation: Mixers in IFAS and digesters will be run at intensities that maintain process performance without over-mixing, thereby reducing avoidable mixing energy.
- Global optimisation: An energy optimisation model will consider all large power consumers together (blowers, pumps, mixers, centrifuges) while ensuring that treatment quality and compliance constraints are always satisfied.

2. IFAS Nitrification Excellence

Objective: Achieve reliable ammonia-N removal with consistent <5 mg/L ammonia at the outlet, including maintaining 75–80% nitrification efficiency during winter..

Key Aspects:

- AI models will explicitly calculate nitrification rates as a function of temperature, DO, SRT, MLSS, and biofilm condition.
- Winter operation will be treated as a separate operating mode, with automatic SRT and DO adjustments to compensate for slower nitrifier growth.
- Biofilm health monitoring will ensure that attached nitrifier populations remain robust and are not inadvertently washed out or stressed.

3. Total Nitrogen Compliance

Objective: Ensure reliable <10 mg/L Total Nitrogen (TN) in the treated effluent through integrated control of nitrification in both suspended sludge and biofilm, and denitrification within biofilm pores and downstream zones.

Approach:

- Nitrification and denitrification models will be coupled to predict how much nitrogen can be removed under current conditions and what adjustments (DO, SRT, RAS, and internal recycle patterns) are needed.
- The system will track the balance between carbon availability, oxygen levels, and anoxic zones in the biofilm to maintain simultaneous nitrification–denitrification.
- When predicted TN approaches the 10 mg/L regulatory limit, the AI will automatically tighten control by increasing nitrification support and enhancing denitrification potential.

4. Chemical Optimisation

Objective: Reduce polymer consumption by 15–25%, $FeCl_3$ usage by 15–20%, and chlorine consumption by 15–25% compared to current fixed-dose practice, without compromising effluent quality or sludge dewatering performance.

Approach:

- Real-time monitoring of sludge characteristics (solids concentration, dewatering behaviour, cake dryness, filtrate clarity) will be used to adjust polymer dose rather than maintaining fixed safety margins.
- Coagulant ($FeCl_3$) dosing will be linked to actual turbidity and solids removal needs, rather than static assumptions, and coordinated with polymer dose to achieve the same effect with less total chemical.
- Chlorine dosing will be based on measured or inferred effluent quality and required CT values, ensuring sufficient disinfection at minimum effective residual.

5. Compliance Assurance

Objective: Increase overall NGT 2019 compliance from the current 85–92% to at least 95% in Year 1, and 99%+ from Year 2 onwards.

Approach:

- Effluent quality prediction models will forecast BOD, TSS, ammonia, and TN several hours ahead using inlet conditions, current process states, and historical patterns.
- When a future risk of non-compliance is detected (e.g., TN predicted to exceed 15 mg/L in 12 hours), the AI will automatically adjust key operating parameters (DO set-points, SRT, RAS/WAS, chemical doses) to bring the system back into a safe operating envelope.
- All critical compliance-related decisions and their rationale will be logged and visible to operators to maintain transparency and trust.

6. Operational Resilience & Predictive Maintenance

Objective: Reduce emergency equipment failures by 40–60% through predictive maintenance and integrated health monitoring for all critical rotating equipment.

Approach:

- Continuous tracking of power, current, vibration, and operating hours will be used to detect patterns indicating bearing wear, misalignment, imbalance, or electrical issues.
- Predictive models will generate early warnings (2–4 weeks) before failure thresholds are reached, allowing planned maintenance to be scheduled.
- Maintenance priorities will be linked to process criticality (e.g., blowers and RAS pumps ranked higher than standby units), ensuring that operational risk is minimised.

7. Sludge Management & Biosolids Quality (Class B to Class A Transition)

Objective: Stabilise and optimise sludge handling to prevent bulking, support digester performance, and progressively upgrade biosolids quality from Class B to Class A over the medium term.

Current Situation:
The plant presently produces Class B biosolids, which have restricted reuse options and lower market or utility value. Coagulant usage and process conditions contribute to higher phosphorus and inert content, making it more difficult to meet tighter Class A criteria.

AI-Driven Improvement Path:

- Bulking prevention: Advanced models will monitor F/M ratio, SRT, DO, temperature, and sludge characteristics to predict bulking risk days in advance and recommend corrective actions (e.g., SRT adjustment, aeration changes) before clarifier performance is affected.
- Digester optimisation: Digester feed rate, mixing intensity, and temperature will be tuned to improve volatile solids destruction and biogas production, leading to better stabilisation of sludge.
- Chemical phosphorus management: Coagulant dosing will be optimised not only for immediate solids removal but also for long-term impact on biosolids phosphorus content. The goal is to gradually reduce chemical phosphorus levels from current >2 mg/L toward more favourable values compatible with Class A targets.
- Pathway to Class A: By combining better stabilisation (through digestion optimisation), improved dewatering, and reduced chemical loading in biosolids, the plant will move progressively towards Class A characteristics. While full reclassification also depends on regulatory testing and specific local standards, the AI system will be explicitly configured to support this transition.

8. Predictive Performance & Operator Decision Support

Objective: Provide operators and management with a clear, predictive view of plant performance, backed by explainable AI recommendations.

Approach:

- Dashboards will display not just current values, but forecast curves for key parameters
- Each AI recommendation (e.g., “reduce WAS by 10%” or “decrease $FeCl_3$ dose by 15%”) will be accompanied by a plain-language explanation and indication of expected impact.
- This ensures that human operators remain in control, understand why the system recommends changes, and can override or refine decisions as necessary.

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5. SOLUTION ARCHITECTURE: AI/ML MODELS

Overview & Flexibility

The AI-driven optimisation system is built around a collection of specialised machine learning models, each designed to address a specific operational challenge or process control requirement. This document describes 20 core models that have been identified through initial assessment and project planning as the foundation for transforming Pirana STP operations.

Important clarification:

These 20 models represent the currently estimated baseline architecture based on our understanding of the plant's processes, challenges, and opportunities. However, the solution is not limited or fixed to exactly these 20 models. During implementation, particularly in Phase 1 (Design & Foundation) and Phase 3 (Core Model Deployment & Testing), the Logicrest team will:

- Conduct detailed site assessments, historical data analysis, and operator interviews.
- Validate which models deliver the highest operational and financial impact.
- Refine, merge, split, or add models based on real plant behaviour, sensor availability, and feasibility.

- Adjust the model architecture to reflect actual conditions discovered during commissioning.

For example:

- If certain process interactions prove simpler than expected, two models might be combined into one.
- If unforeseen operational patterns are discovered (e.g., specific shock load signatures unique to Ahmedabad's drainage network), additional specialised models may be introduced.
- If certain sensors prove unreliable or unavailable, some models may be adapted to work with alternative data sources.

The commitment is not to deliver exactly 20 fixed models; it is to deliver a complete, effective, and optimised AI control system that achieves all the performance objectives outlined in Section 4.

This flexibility ensures that the solution remains practical, cost-effective, and aligned with real-world operational needs rather than being constrained by theoretical design assumptions.

TIER 1: CORE Optimisation MODELS (Models 1-8)

Foundation models that handle basic process control (aeration, chemical dosing, energy, equipment health, anomaly detection). These are essential for any well-functioning STP and form the baseline AI control layer.

Model 1: DO Set-Point Optimisation

Purpose in simpler terms
This model decides how much oxygen should be kept in the aeration tanks at any moment. It acts like a smart “breathing controller” for the plant, so the microorganisms always have enough oxygen to clean the water, but not so much that energy is wasted or the IFAS biofilm is stressed.

What problem this solves
- Today, operators use mostly fixed DO set-points (for example 3–4 mg/L) for the whole day and seasons, even when the incoming load is low. This wastes a large amount of blower energy.
- IFAS biofilm on the media cages needs stable and appropriate oxygen conditions. Large and frequent DO swings can make the biofilm unstable, cause parts of it to detach, and reduce

nitrification performance.
- Ammonia removal is very sensitive to DO. Manual and slow adjustments lead to periods of under-aeration (ammonia breakthrough risk) and over-aeration (unnecessary power consumption).

What inputs this model uses
- Dissolved Oxygen readings from multiple sensors in the aeration tanks.
- Incoming load data: BOD/COD, ammonia concentration, flow, temperature, pH.
- MLSS (mixed liquor suspended solids) concentration in the aeration tanks.
- Ammonia concentration at the aeration outlet.
- Blower air flow and power consumption (to understand oxygen transfer efficiency).
- Indirect indicators of biofilm condition around IFAS cages, such as how quickly DO drops "around the cages" (oxygen consumption in that zone) and trends in oxygen transfer efficiency.

How the model works
- It first estimates the **instantaneous oxygen demand** by looking at how much organic matter and ammonia is coming in, how much biomass is present, and what the current temperature is (colder water means bacteria work more slowly and need more time and therefore more oxygen overall).
- It then compares DO readings in different parts of the tank, especially close to IFAS media zones versus open water. From this, it infers how intensely oxygen is being consumed around the cages – this "mass around cages" behaviour gives a picture of how active and dense the biofilm is.
- By tracking oxygen transfer efficiency over days and weeks, the model detects if the biofilm layer is becoming too thick. A thick film consumes a lot of oxygen at the surface and starves the inner layers. The model is configured to keep the **effective active film thickness within a healthy band (approximately 1–5 mm)** by avoiding oxygen conditions that allow uncontrolled thickening.
- For each possible DO set-point (for example 2.2, 2.4, 2.6, 2.8 mg/L), the model estimates:
- Expected BOD and ammonia removal.
- Risk of non-compliance.
- Blower effort required.
- It then selects the **lowest DO set-point that still guarantees reliable treatment and biofilm health**.
- The calculation is repeated automatically every few minutes. DO targets are adjusted gently and continuously rather than with large, infrequent manual changes.

Outputs and integration with other models
- Main output: a continuously updated **DO set-point** for aeration control (typically within a

range like 2.2–2.8 mg/L under normal conditions).
- It sends this target to the aeration control system, which then adjusts blower speeds and valves.
- It provides a **biofilm health indicator** to other models, summarising whether oxygen conditions around the media are favourable.
- It works closely with:
- **Energy Optimisation (Model 3):** to meet DO targets using minimum blower power.
- **Nitrification Prediction (Model 9):** to ensure the oxygen level supports the required ammonia removal.
- **Bulking / Filament Risk (Model 15):** because poor DO control increases bulking risk.
- **Aeration Demand Forecasting (Model 2):** which informs Model 1 about likely upcoming changes in load.

Real-world example
At night, the inflow and pollution load are low. The model sees that less oxygen is required and that the IFAS biofilm is stable. It gradually reduces the DO set-point from 2.7 mg/L to 2.3 mg/L. Blowers slow down, saving energy, while effluent quality remains within limits and the biofilm remains healthy.
In the morning, the inflow and ammonia load start to climb. The model detects this from rising flow and inlet analysers and predicts higher oxygen demand. It gently increases the DO set-point to around 2.7–2.8 mg/L before a problem appears, so there is no ammonia spike and no need for emergency manual intervention.

Model 2: Aeration Demand Forecasting

Purpose in simpler terms
This model looks into the near future (the next few hours) and predicts how much aeration the plant will need. It is similar to a “weather forecast” for the biological process, so the plant can prepare blowers and oxygen supply in advance instead of reacting late.

What problem this solves
- Without forecasting, operators only react after DO drops or ammonia increases. This lag can cause short-term non-compliant conditions or high stress on equipment.
- Daily patterns (morning peaks, night lows), weekly differences (weekdays versus weekends), and seasonal effects (summer versus winter) all change the aeration requirement, but these are not systematically used in current operation.
- Sudden changes like rain events or industrial discharges create shock loads. If the plant does not anticipate these, aeration is often adjusted late and in a less controlled manner.

What inputs this model uses

- Historical records of influent flow and load: typical 24‑hour patterns, weekday/weekend differences, seasonal variations.
- Real‑time influent data: flow, BOD/COD, ammonia.
- Real‑time and forecast wastewater temperature (and, if available, local weather).
- Output from the IFAS‑related models (especially nitrification performance), to understand how sensitive the current system is to load and temperature changes.
- Calendar information (e.g. weekends, festivals) which often change city sewage patterns.

How the model works

- It learns from past data how the plant typically behaves: for example, that on working days flow usually starts rising around 6–7 AM, peaks mid‑morning, falls in the afternoon, and dips at night.
- It then compares current conditions to these patterns. If today's morning flow is already higher than usual, it adjusts the afternoon prediction upwards; if a storm is forecast, it expects higher flow with more dilution.
- The model also includes the effect of temperature on nitrifying bacteria. When temperature is predicted to go down (e.g. winter evenings), it assumes nitrification will slow and that more aeration effort will be required for the same ammonia removal.
- For each time horizon (for example +1 hour, +2 hours, +4 hours, +8 hours), it estimates the **percentage of maximum aeration capacity likely to be needed** and attaches a confidence level.
- Based on this, it can recommend when to stage additional blowers gradually, and when it is safe to take blowers offline or reduce speeds.

Outputs and integration with other models

- Main output: a **time‑series forecast of aeration demand**, such as "in 2 hours, expected demand is 90–95% of current capacity; in 8 hours (night), expected demand is 50–60%".
- These forecasts are used by:
- **Model 1 (DO Set‑Point):** to adjust DO targets a little earlier before load changes actually reach the aeration tanks.
- **Model 3 (Energy Optimisation):** to plan blower staging, pump operation and night‑time energy saving modes without compromising treatment.
- **Model 14 (Seasonal SRT Adjustment):** to coordinate sludge age strategy with changing temperature and aeration demand.
- Operators see these forecasts as simple graphs and recommended actions ("prepare to start an additional blower in 45 minutes").

Real‑world example

At 6 AM, the model predicts that influent flow will rise sharply between 7 AM and 9 AM based

on historical data and current readings. It also sees that wastewater temperature is slightly lower than usual, meaning nitrifiers will work more slowly. It forecasts that aeration demand will reach around 95% of capacity by 8:30 AM.
The system then advises (and can automatically enforce) that a standby blower be brought online now and ramped up slowly. As a result, when the actual morning peak arrives, there is already enough aeration in place, DO remains stable, and ammonia removal remains strong without emergency manual action.

Model 3: Energy Optimisation

Purpose in simpler terms
This model is the "energy manager" for the entire treatment plant. It looks at all the major electrical consumers—blowers, pumps, mixers, and dewatering equipment—and decides how to run them together in the most efficient way, while still fully meeting all treatment and regulatory targets.

What problem this solves
- Blowers, pumps and mixers are often run in a conservative, fixed way (for example, several units running at moderate speed all the time), which is safe but not efficient.
- Each system (aeration, pumping, mixing) is normally adjusted separately. There is no single control layer that asks, "What is the best overall combination of all equipment settings for the plant as a whole?"
- Without a coordinated view, it is easy to over-aerate, over-pump, or over-mix, especially during low-demand periods like nights and weekends.

What inputs this model uses
- Real-time power and status measurements from all major motors and drives (blowers, influent/RAS/WAS/effluent pumps, mixers, centrifuges).
- Treatment performance indicators: BOD, TSS, ammonia, total nitrogen at key points, MLSS levels, sludge blanket levels, digester status.
- Equipment characteristics: efficiency curves for motors and pumps, minimum and maximum speeds, recommended operating ranges, allowed start/stop frequency.
- Forecasts from other models:
- Aeration demand from Model 2.
- DO targets from Model 1.
- Predicted effluent quality from Model 5.
- Biological state from Model 13 (MLSS-SRT-F/M balance).

How the model works

- It defines the **non-negotiable limits** for the plant: effluent BOD, TSS and TN must stay within discharge standards; MLSS and SRT must stay within safe ranges; DO must not fall so low that treatment is at risk.
- It then considers all controllable equipment: which blowers are on, what speeds they run at; which pumps are running and at what flow; what speeds mixers operate at; when centrifuges run.
- For each small time window (for example, every 10–15 minutes), it evaluates many combinations of these equipment settings and estimates:
- Whether all treatment constraints will still be satisfied.
- How much electrical energy each combination would consume.
- It chooses the combination that **meets all process constraints with the lowest overall energy use**.
- Typical strategies it uses include:
- Running fewer blowers or pumps at their most efficient speed instead of more units at inefficient low loads.
- Reducing mixer intensity or switching to intermittent mixing during stable, low-load periods.
- Staggering large motor start/stop events to avoid sudden large power changes and to reduce mechanical stress.

Outputs and integration with other models

- Main outputs: **speed and on/off commands** for blower VFDs, pump VFDs, and mixer VFDs, and scheduling recommendations for dewatering equipment.
- It uses DO targets from Model 1 as a hard requirement and finds the most efficient way for blowers to achieve those DO levels.
- It uses predictions from Model 2 and Model 5 to decide whether it is safe to reduce energy use further or whether a period of higher demand is coming.
- It coordinates with Model 13 to ensure that any energy-saving actions do not push MLSS, SRT or F/M outside their optimal ranges.
- Operators see the effect as more stable, smoother operation, with equipment working in a well-planned pattern instead of sudden manual changes.

Real-world example

During the night, when flows and loads are low, the model sees that only one blower running at around 60% speed is enough to maintain the DO set-point from Model 1 and keep treatment fully compliant. It therefore parks the other blowers and reduces mixer speeds slightly, while ensuring MLSS and clarifier performance remain stable.
As the morning peak approaches, it uses the forecast from Model 2 to bring a second blower online early and ramp it up gradually, instead of waiting until DO drops. Throughout the day, it

continuously readjusts blower speeds, pump combinations, and mixer intensities so that the plant always runs at the least-energy configuration that still satisfies all process and compliance constraints.

Model 4: Chemical Dosing Optimisation

Purpose in simpler terms

This model is the “smart dosing assistant” for the plant’s main treatment chemicals: polymer (for sludge dewatering), ferric chloride / coagulant (for solids and phosphorus removal), and chlorine (for final disinfection). It adjusts chemical doses continuously based on what is actually happening in the plant, instead of relying on fixed set-points with large safety margins.

What problem this solves

- Polymer for dewatering is usually set at a fixed rate based on past experience, even though sludge properties change daily. This can lead to using more chemical than actually needed or, on bad days, too little polymer and poor dewatering performance.
- Ferric chloride is often dosed as a constant rate based on “typical conditions”, without considering real-time turbidity, solids load, or phosphorus needs, and without looking at the long-term impact on biosolids quality.
- Chlorine is frequently added at a fixed dose designed to handle worst-case conditions, even when the effluent is very clean and needs much less disinfectant to achieve the same level of safety.
- Overall, this fixed-dose approach wastes chemicals and also reduces the ability to control biosolids quality (for example, high chemical phosphorus content which prevents reaching Class A biosolids).

What inputs this model uses

For polymer dosing (sludge dewatering):

- **Total solids after the aeration tank and before the SCL point** (a key measurement): how concentrated the sludge is after biological treatment and initial thickening, but before final sludge concentration and centrifuge feed.
- Sludge flow rate to the dewatering equipment.
- Sludge temperature and sludge age (SRT), which influence how easily sludge can be dewatered.
- Observed centrifuge performance: cake dryness and filtrate clarity.

For ferric chloride / coagulant dosing:

- Influent turbidity and TSS (how much suspended material needs to be captured).

- Measured or estimated phosphorus levels.
- pH and alkalinity, which influence how ferric chloride behaves chemically.
- Sludge production rate and clarifier performance.

For chlorine dosing (disinfection):
- Effluent BOD and TSS (more organic matter and solids mean more chlorine demand).
- Effluent flow rate and the resulting contact time in the chlorine contact tank.
- Measured chlorine residual at the outlet of the contact tank.
- Microbiological target: required disinfection level expressed as fecal coliform limit.

How the model works

Polymer (dewatering) control:
- The model continuously watches the **total solids after aeration and before the SCL**. If this percentage is high, it knows the sludge is already well concentrated and less polymer per kilogram of solids should be needed. If the solids concentration falls, the sludge will be more dilute and usually harder to dewater, so more polymer may be needed.
- Using historical data, it learns the relationship between:
- Solids concentration, temperature, and SRT, and
- The polymer dose required to reliably reach the target cake dryness.
- From this, it calculates an optimal polymer dose (for example, kg of polymer per ton of dry solids) for the current conditions, then translates that into a dosing pump speed.
- It monitors the resulting cake dryness and filtrate clarity and makes small corrections up or down, learning over time what works best for the plant's specific sludge.

Ferric chloride / coagulant control:
- The model estimates how much coagulant is actually required based on real-time turbidity, suspended solids and phosphorus removal needs.
- It also considers how polymer and coagulant work together. Very often, a slightly lower coagulant dose combined with a slightly adjusted polymer dose can achieve the same or better clarification and thickening performance, but with less total chemical.
- Additionally, it takes into account the long-term effect on biosolids. Because ferric chloride adds iron and binds phosphorus into the sludge, the model aims to use only as much as needed, helping to gradually reduce chemical phosphorus levels in the biosolids over months and improve the pathway towards Class A biosolids.

Chlorine (disinfection) control:
- The model calculates the required **CT (Concentration × Time)** value for disinfection, using the current effluent quality, flow and contact tank volume. If effluent is very clear and low in BOD/TSS, the required chlorine concentration is lower; if effluent quality is more challenging or

contact time is shorter due to high flow, then a higher concentration is needed.
- It sets the chlorine dose so that, with the actual contact time, the desired disinfection is achieved while also meeting a target residual at the outlet of the contact tank.
- The actual measured residual is fed back to the model, which then fine-tunes the dose: if residual is consistently higher than needed, it slightly reduces dose; if residual is too low, it increases dose.

Outputs and integration with other models
- Outputs:
- Recommended dosing rates (pump speeds or flow set-points) for:
- Polymer dosing pumps.
- Ferric chloride / coagulant dosing pumps.
- Chlorine dosing system.
- Integration:
- Works with **Model 3 (Energy Optimisation)** because better dewatering and proper chemical dosing can reduce the effort required by centrifuges and pumps.
- Uses information from **Model 13 (MLSS-SRT-F/M)** and **Model 15 (Bulking Risk)**, since changes in sludge characteristics affect polymer needs.
- Supports the **biosolids quality objective** from the Project Aim section by actively managing coagulant dosing to reduce long-term chemical phosphorus in sludge.

Real-world example
On a day when sludge settles and thickens very well, the total solids after aeration and before SCL rise to a high value. Model 4 observes that the sludge going to the centrifuge is already fairly concentrated and, based on past behaviour, will achieve the target cake dryness with less polymer. It reduces the polymer dosing rate slightly. The centrifuge still produces cake with the desired dryness and clean filtrate, confirming that the lower dose is sufficient.
On another day, rain dilutes the sludge, and solids concentration at the same point drops. The model detects this and predicts that the sludge will now be harder to dewater. It increases the polymer dose automatically and the plant continues to achieve stable dewatering performance without operators having to manually trial new set-points. At the same time, the model continuously adjusts ferric chloride and chlorine rates based on real-time solids and effluent quality, ensuring that clarification, phosphorus removal, and disinfection all remain within target ranges without unnecessary overdosing.

Model 5: Effluent Quality Prediction

Purpose in simpler terms
This model forecasts the quality of treated water leaving the plant hours or even a day in advance. It predicts key parameters like BOD, TSS, ammonia, and Total Nitrogen before they are

actually measured, giving operators early warning if the plant is drifting toward a compliance problem.

What problem this solves

- Today, operators only know effluent quality after laboratory tests or when online analyzers show results, which reflect what already happened hours ago.
- By the time a problem is detected (for example, ammonia rising above target), it is often too late to prevent a violation, and corrective actions take additional hours to show effect.
- There is no systematic way to see "where the plant is heading" or to catch gradual deterioration early.
- Winter periods and shock load events often surprise operators because there is no predictive tool showing how current conditions will affect effluent quality later.

What inputs this model uses

- Current and recent influent characteristics: BOD, COD, ammonia, flow, temperature.
- Current aeration tank conditions: DO, MLSS, SRT, temperature, mixing patterns.
- IFAS biofilm health and activity status from Model 11.
- Nitrification and denitrification rates predicted by Models 9 and 10.
- Recent trends in effluent quality (the model learns typical "lag times" between changes in the aeration tanks and changes at the outlet).
- Weather and seasonal information (temperature trends, rainfall forecasts).

How the model works

- The model uses machine learning trained on months or years of historical data from the plant. It learns patterns such as:
- "When influent ammonia increases by X mg/L and temperature is Y°C, effluent ammonia typically rises by Z mg/L after approximately 8–12 hours."
- "When MLSS is maintained at A mg/L and DO at B mg/L under C°C conditions, effluent BOD is reliably below 5 mg/L."
- It combines the current state of the biological process with predicted changes (such as incoming load forecasts from Model 2, or temperature trends) to estimate what effluent quality will look like at specific future times.
- It generates forecasts for multiple horizons (for example, +4 hours, +8 hours, +12 hours, +24 hours) and assigns confidence levels to each prediction.
- If a prediction shows that a parameter (especially Total Nitrogen or ammonia) is likely to approach or exceed the regulatory limit, the model raises an alert.

Outputs and integration with other models

- Main outputs:
- Predicted effluent BOD, TSS, ammonia, and Total Nitrogen for the next 4, 8, 12, and 24 hours,

with confidence intervals.
- Risk indicators: for example, “80% probability that TN will be 13–15 mg/L in 12 hours” or “High confidence effluent will remain comfortably compliant for next 24 hours.”
- Integration:
- Feeds into **Model 7 (Operator Advisory)** to generate alerts and recommendations when predicted quality approaches limits.
- Informs **Model 3 (Energy Optimisation)**: if effluent quality is predicted to have a comfortable safety margin, energy use can be optimised more aggressively; if quality is marginal, operate more conservatively.
- Triggers adjustments in **Model 13 (MLSS-SRT-F/M)** and **Model 14 (Seasonal SRT)** to proactively change biological conditions before a violation occurs.
- Receives inputs from **Models 1, 2, 9, and 10** to understand aeration, load trends, and nitrogen removal performance.

Real-world example
On a Tuesday afternoon, the model detects that wastewater temperature has dropped by 4°C overnight and nitrification efficiency is beginning to decline (ammonia removal slowing down). It predicts that if current conditions continue without adjustment, effluent Total Nitrogen will rise from the current 8 mg/L to approximately 10 mg/L by Thursday morning—approaching the 10 mg/L regulatory limit.
The system raises an alert 36 hours in advance. Based on this warning, Model 14 automatically begins increasing SRT (reducing waste sludge flow), and Model 1 raises the DO set-point slightly. By Thursday morning, when the predicted impact would have occurred, the plant has already compensated and actual effluent TN remains at 8.5 mg/L, well within compliance. The violation is prevented proactively instead of reacted to after it happens.

Model 6: Equipment Health Monitoring & Predictive Maintenance

Purpose in simpler terms
This model continuously watches the condition and performance of all major rotating equipment (blowers, pumps, mixers, centrifuges) and predicts when failures are likely to occur, typically 2–4 weeks before actual breakdown. This allows maintenance to be planned during convenient low-demand periods instead of dealing with emergency failures.

What problem this solves
- Equipment failures are currently discovered only when a machine stops working or shows obvious signs of distress (loud noise, overheating, complete shutdown).
- Emergency repairs are disruptive, often require expensive expedited parts and contractor

response, and can compromise treatment capacity or compliance during the repair period.
- There is no systematic way to track the gradual degradation that occurs in motors, bearings, and pumps over weeks and months before failure.
- Maintenance is either reactive (fix it after it breaks) or time-based (replace parts on a schedule), neither of which is optimal compared to condition-based maintenance.

What inputs this model uses
- Real-time electrical parameters from all motors: current draw (amperes), voltage, power factor, total power consumption.
- Vibration data (if vibration sensors are installed) or inferred vibration patterns from power fluctuations and motor behaviour.
- Operating hours and duty cycles: how many hours each piece of equipment has run, how many start/stop cycles, how long at high load versus low load.
- Performance efficiency trends: for example, a blower that used to deliver a certain air flow at a certain power consumption but now requires more power for the same output.
- Temperature monitoring: motor winding temperature or bearing temperature, if available.

How the model works
- The model first establishes a **baseline performance signature** for each piece of equipment when it is healthy. For example:
- “Blower #2 normally consumes 145 kW at 75% speed and delivers 8,500 m^3/hr of air.”
- “RAS Pump #3 normally draws 42 amps at rated flow and shows smooth power with minimal fluctuation.”
- It then continuously monitors for signs of gradual degradation:
- **Increasing power consumption for the same work:** Often indicates bearing wear, motor winding degradation, or mechanical friction increasing.
- **Vibration patterns changing:** Developing bearing defects or mechanical imbalance show up as increasing or changing vibration frequencies.
- **Efficiency declining:** A pump that now uses 10–15% more energy to move the same volume of water is likely experiencing wear.
- **Temperature rising:** Overheating bearings or motor windings indicate problems developing.
- When the model detects a concerning trend, it estimates how quickly the degradation is progressing and predicts when the equipment will reach a failure threshold.
- It generates alerts ranked by criticality: for example, blowers and RAS pumps (critical for continuous operation) are flagged as higher priority than standby units or less-critical equipment.

Outputs and integration with other models
- Main outputs:

- **Equipment health scores** (0–100%) for each major motor and pump.
- **Predicted failure dates** for equipment showing concerning trends (for example, “Blower #2 bearing failure likely in 18–25 days”).
- **Maintenance recommendations**: “Schedule bearing replacement for Blower #2 during next low-flow weekend.”
- Integration:
- Feeds into **Model 7 (Operator Advisory)** to display equipment status and maintenance alerts on dashboards.
- Coordinates with **Model 3 (Energy Optimisation)** to avoid heavy reliance on equipment showing early signs of distress, and to rotate load to healthier units.
- Supports the plant’s maintenance management system with advance scheduling information.

Real-world example
In Week 1, the model notices that Blower #2’s power consumption has increased by approximately 6% compared to baseline, even though its output (air flow) has not increased. By Week 3, the increase is 8% and there are small changes in the vibration signature detected from power fluctuations. The model analyses this pattern and recognises it as consistent with bearing lubrication degrading.
By Week 4, it issues an alert: “Blower #2 bearing failure predicted in 2–3 weeks; recommend inspection and bearing replacement during next scheduled low-flow period.” Maintenance staff schedule the work for the coming weekend when flows are low and treatment demand is easily handled by the other blowers. The bearing is replaced during planned downtime, and Blower #2 returns to service with restored efficiency. No emergency breakdown occurs, no compliance risk, and no expensive emergency contractor call-out.

Model 7: Operator Advisory System

Purpose in simpler terms
This model is the “intelligent assistant” for plant operators. It takes information from all the other AI models and translates it into clear, actionable recommendations displayed on operator screens. It also explains why each recommendation is being made, so operators understand the reasoning and can make informed decisions.

What problem this solves
- Currently, treatment success depends heavily on individual operator experience and skill. Different operators may make different decisions in the same situation, leading to variable plant performance.
- Operators are often hesitant to trust “black box” AI systems that give instructions without

explaining why. This creates resistance to automation.
- When multiple things are happening at once (for example, rising influent load, falling temperature, and equipment maintenance), it is difficult for a single person to prioritise and coordinate all the necessary responses.
- Senior operator knowledge is not systematically captured. When experienced staff retire or leave, their expertise is lost.

What inputs this model uses
- All real-time process data: DO, MLSS, flows, loads, temperatures, equipment status, and effluent quality.
- Outputs and recommendations from every other AI model (Models 1–6, 8–20).
- Current compliance status and risk indicators (is the plant operating with comfortable margins or close to limits?).
- Historical records of operator actions and outcomes (used to learn and refine recommendations).

How the model works
- Model 7 acts as a “synthesis layer” that combines information from all other models into a unified operator interface.
- It prioritises information by urgency and importance:
- **Green status**: All systems operating well within normal ranges; no action needed.
- **Yellow status**: Attention recommended; a parameter is trending toward a limit or an optimisation opportunity exists.
- **Red status**: Urgent action required; compliance risk or equipment issue detected.
- For each recommendation (for example, “Reduce WAS flow by 10%” or “Increase DO set-point to 2.7 mg/L”), the model provides a **plain-language explanation** of the reasoning. For example:
- “Temperature has dropped to 13°C overnight. Nitrifying bacteria grow more slowly in cold water. Reducing WAS gives them more time in the system (higher SRT) to establish the population needed for ammonia removal.”
- “Effluent Total Nitrogen is predicted to reach 14 mg/L in 12 hours. Raising DO set-point will improve nitrification efficiency and bring TN back below 13 mg/L.”
- The model uses **explainable AI (XAI)** techniques, specifically SHAP (SHapley Additive exPlanations), to show which factors are most influential in each recommendation. Operators can see, for example, “This recommendation is primarily driven by temperature (60% influence), secondarily by current SRT (25%), and also considers recent ammonia trends (15%).”
- Operators always retain the ability to accept, modify, or reject recommendations. The system learns from operator feedback.

Outputs and integration with other models
- Main outputs:
- **Process status dashboard**: Visual display showing current state of all major systems with colour-coded health indicators.
- **Prioritised recommendation list**: Top 3–5 actions operators should consider, ranked by urgency and impact.
- **Performance forecasts**: Graphs showing predicted DO, MLSS, effluent quality for the next 12–24 hours.
- **Compliance alerts**: Clear warnings if any parameter is predicted to approach or exceed regulatory limits.
- **Plain-language explanations**: For every recommendation, a short text explanation of why it is suggested and what outcome is expected.
- Integration:
- Receives inputs from **all 20 models** and synthesises them into unified guidance.
- Feeds back operator decisions and outcomes to improve future recommendations (continuous learning).
- Displays information from **Model 5** (effluent quality predictions), **Model 6** (equipment health), **Model 9 and 10** (nitrification and denitrification performance), and others in a coordinated way.

Real-world example
At 10 AM, the dashboard shows a yellow alert: “Nitrification performance declining – Total Nitrogen predicted to reach 9.5 mg/L in 12 hours.”
The operator clicks on the alert and sees the recommendation: “Increase SRT by reducing WAS flow from 45 m^3/hr to 38 m^3/hr.”
The explanation reads: “Temperature dropped to 13°C overnight. Nitrifying bacteria grow approximately 40% slower at this temperature compared to summer conditions. Reducing WAS flow increases the average time bacteria spend in the system (SRT increases from 16 days to approximately 18 days), allowing nitrifiers to build up a larger population despite slower growth. Expected result: TN stabilises at 8–9 mg/L within 24 hours.”
The operator reviews the explanation, agrees with the logic, and accepts the recommendation. WAS flow is reduced, and over the next day, ammonia and Total Nitrogen remain within comfortable compliance margins. The operator learns the reasoning and will be better prepared to handle similar situations in future.

Model 8: Anomaly Detection

Purpose in simpler terms
This model continuously watches all sensor data and operational patterns across the plant and

identifies unusual events or conditions that do not match normal behaviour. It acts as an “early warning radar” for shock loads, equipment malfunctions, sensor failures, toxic discharges, or other unexpected problems.

What problem this solves

- Unusual events (shock loads from industrial discharges, sudden changes in influent characteristics, equipment beginning to malfunction) are often not noticed until they cause visible problems like effluent quality deterioration or equipment alarms.
- Traditional alarm systems trigger on simple thresholds (for example, “DO below 2.0 mg/L”) which often create false alarms during normal transient conditions, leading to “alarm fatigue” where operators start ignoring alerts.
- It is difficult for operators to distinguish between normal process variability (which happens every day) and genuine anomalies that require investigation.
- By the time a problem is clearly visible, it may already have caused treatment upset or equipment damage.

What inputs this model uses

- Real-time data streams from all sensors across the plant (typically 30+ measurement points): influent flow and quality, aeration tank conditions, clarifier levels, effluent quality, equipment power consumption, temperatures, pressures, etc.
- Historical “normal operation” data: the model learns what typical patterns look like (daily cycles, weekly patterns, seasonal variations).
- Expected ranges for each parameter based on time of day, day of week, season, and current operating mode.
- Correlation patterns: for example, “When influent flow increases, DO typically decreases slightly; if DO drops sharply without a flow change, that is anomalous.”

How the model works

- The model first builds a **statistical picture of normal operation** by analysing months of historical data. It learns, for example:
- “Influent flow on weekdays between 8 AM and 10 AM is typically 180–220 m^3/hr.”
- “DO in Aeration Tank 2 normally varies between 2.3–2.9 mg/L with slow, smooth changes.”
- “Blower power consumption is strongly correlated with DO set-point and flow; deviations from this pattern indicate something unusual.”
- It then continuously compares current measurements to these learned patterns using machine learning techniques (such as isolation forests, statistical control charts, or auto-encoders).
- When it detects something that does not fit the normal pattern, it evaluates:
- **How unusual is it?** (Slightly outside normal range, or very far outside?)
- **How sudden is the change?** (Gradual drift versus abrupt spike?)

- **Are multiple related parameters also showing anomalies?** (For example, influent COD spike plus pH drop = likely industrial discharge; single sensor anomaly = possible sensor fault.)
- It classifies detected anomalies by type:
- **Shock load or toxic discharge**: Sudden change in influent quality (COD spike, pH change, unusual chemical detected).
- **Equipment malfunction**: Power consumption pattern change, vibration increase, efficiency drop.
- **Sensor fault**: Single sensor reading inconsistent with all related measurements.
- **Process upset**: Biological indicators (MLSS, sludge settling, ammonia removal) deteriorating in a pattern consistent with bulking, nitrification inhibition, or other biological stress.
- For each detected anomaly, the model generates an alert with:
- Description of what is unusual.
- Suggested root cause (for example, “Pattern consistent with industrial discharge from upstream”).
- Recommended immediate actions (for example, “Monitor DO and ammonia closely; increase aeration if needed; check for influent pH or toxic compounds”).

Outputs and integration with other models
- Main outputs:
- **Anomaly alerts** with severity classification (low, medium, high).
- **Root cause suggestions** based on pattern recognition.
- **Recommended actions** to investigate or mitigate the anomaly.
- **Trend graphs** showing which parameters are deviating and how quickly.
- Integration:
- Feeds into **Model 7 (Operator Advisory)** to display alerts prominently on dashboards.
- Coordinates with **Model 6 (Equipment Health Monitoring)** to cross-check whether an anomaly is due to equipment degradation.
- Informs **Model 2 (Aeration Demand Forecasting)** and **Model 5 (Effluent Quality Prediction)** when shock loads are detected, so forecasts can be adjusted accordingly.
- Can trigger automatic protective responses: for example, if a severe toxic discharge is detected, the system could recommend temporarily increasing DO and reducing WAS to protect the biological process.

Real-world example
On a Wednesday afternoon, the model detects that influent COD has spiked from a typical 320 mg/L to 580 mg/L within 30 minutes, and simultaneously influent pH has dropped from 7.2 to 6.4. No corresponding change in flow is observed.
The model recognises this pattern as consistent with an industrial discharge event (high organic load plus acidic wastewater). It immediately generates a high-priority alert: “Anomaly detected

– Influent COD spike + pH drop likely indicates industrial discharge."
The recommended actions displayed to operators are:
- "Monitor aeration tank DO closely; oxygen demand will increase significantly in the next 2–3 hours as this high-COD slug reaches the biological process."
- "Prepare to stage additional blowers."
- "Monitor pH in aeration tanks; if pH drops below 6.5, consider alkalinity addition to prevent nitrification inhibition."
- "Check upstream industrial connections for unauthorised discharge."
Operators follow these recommendations. Blowers are staged early, DO remains stable, and the high-load slug is treated successfully without causing an upset or compliance problem. Meanwhile, plant management contacts upstream industries to investigate and prevent recurrence.

TIER 2: IFAS-SPECIFIC MODULES (Models 9-14) - ESSENTIAL FOR COMPLIANCE

Advanced models designed explicitly for the IFAS upgrade—nitrification, denitrification, biofilm health, seasonal adjustments. These unlock the full potential of the biofilm-based treatment system.

Model 9: Nitrification Rate Prediction

Purpose in simpler terms
This model predicts how quickly ammonia will be converted to nitrate by nitrifying bacteria. Nitrification is a critical process for nitrogen removal and is highly sensitive to temperature, oxygen levels, and the health of both suspended bacteria and the IFAS biofilm. This model forecasts nitrification performance hours ahead so the plant can adjust conditions before ammonia removal fails.

What problem this solves
- Nitrification efficiency varies greatly with temperature. In winter (10–15°C), nitrifying bacteria work 3–5 times slower than in summer (25–30°C). Without prediction, winter nitrification failures happen every year and are only noticed after ammonia levels have already risen.
- Operators cannot easily tell whether current conditions will maintain nitrification or whether ammonia breakthrough is developing until it appears in effluent measurements (by which time it is too late to prevent).
- The IFAS system has two populations of nitrifiers: suspended in the mixed liquor and attached to the biofilm. Current operation does not explicitly account for the contribution of each, or optimise conditions for both simultaneously.

What inputs this model uses
- **Ammonia concentration** in the aeration tank, measured by real-time online analyser.
- **Dissolved oxygen** from multiple sensors across aeration zones (nitrifiers need adequate oxygen to function).
- **Temperature** of the wastewater (this is the single most important factor affecting nitrification rate).
- **MLSS concentration** (indicates how much suspended biomass, including nitrifiers, is present).
- **Estimated biofilm mass and health** from Model 11 (biofilm contributes significantly to nitrification in IFAS systems).
- **pH and alkalinity**: Nitrifiers are sensitive to pH; they also consume alkalinity during the nitrification reaction, so low alkalinity can slow or stop nitrification.
- **Sludge age (SRT)**: Nitrifying bacteria grow slowly, so they need sufficient time in the system. Low SRT can "wash out" nitrifiers before they establish a stable population.

How the model works
- The model calculates nitrification rate separately for two pathways:
- **Suspended nitrifiers** in the mixed liquor (MLSS).
- **Attached nitrifiers** in the IFAS biofilm on the media cages.
- For each pathway, it applies temperature correction factors based on established microbiology principles:
- At 25°C (warm conditions): Nitrifiers are at or near maximum activity = 100% baseline rate.
- At 20°C: Nitrification rate approximately 70–80% of baseline.
- At 15°C: Nitrification rate approximately 45–55% of baseline.
- At 10°C: Nitrification rate approximately 25–35% of baseline.
- It also considers DO availability. If DO is below approximately 2.0 mg/L, nitrification rate begins to decline because nitrifiers are oxygen-limited. If DO is above about 2.5 mg/L under normal conditions, further increases in DO provide diminishing returns.
- The biofilm nitrifier contribution is especially important in winter, because biofilm bacteria are somewhat more cold-tolerant and stable than suspended bacteria. The model estimates how much ammonia is being removed by the biofilm versus the suspended phase.
- Combining these factors, the model predicts:
- **Ammonia removal rate** in mg NH_4-N per hour.
- **Outlet ammonia concentration** expected in 4, 8, or 12 hours.
- **Nitrification efficiency** as a percentage (how much of the incoming ammonia load is being successfully converted).
- If predicted outlet ammonia approaches or exceeds the target (typically 5 mg/L as an internal limit), the model generates an alert.

Outputs and integration with other models
- Main outputs:
- **Predicted nitrification rate** (mg NH_4-N removed per hour or per day).
- **Forecast outlet ammonia concentration** for multiple time horizons (for example, "predicted 3.8 mg/L in 8 hours" or "predicted 6.2 mg/L in 24 hours if no action taken").
- **Nitrification efficiency percentage** (for example, "currently 82% efficient; winter target is 75–80% to maintain compliance").
- Integration:
- Feeds into **Model 5 (Effluent Quality Prediction)** because ammonia and Total Nitrogen predictions depend directly on nitrification performance.
- Feeds into **Model 10 (Denitrification Prediction)** because the amount of nitrate produced by nitrification determines how much denitrification is possible.
- Informs **Model 14 (Seasonal SRT Adjustment)**: when nitrification efficiency is predicted to decline due to cold temperature, Model 14 responds by increasing sludge age.
- Receives DO targets from **Model 1** and uses them to estimate whether oxygen supply is adequate for current nitrification needs.
- Receives biofilm health status from **Model 11** to adjust predictions based on whether the IFAS biofilm is performing well or is stressed.

Real-world example
In early January, wastewater temperature drops to 12°C. The model calculates that, at this temperature, the nitrification rate of suspended bacteria is approximately 45% of their summer capacity. However, the IFAS biofilm is still contributing robustly (biofilm bacteria are more cold-stable).
At current operating conditions (SRT = 15 days, DO = 2.5 mg/L), the model predicts that combined nitrification from suspended + biofilm bacteria will remove approximately 75% of incoming ammonia, resulting in an outlet ammonia concentration of approximately 6.8 mg/L—above the internal target of 5 mg/L and creating a compliance risk.
The model issues an alert and recommends:
- "Increase SRT to 22 days by reducing waste sludge flow" (gives nitrifiers more time to build population despite slow growth).
- "Increase DO set-point to 2.8 mg/L" (ensures oxygen is not a limiting factor).
These adjustments are made automatically or with operator approval. Over the following days, the nitrifier population stabilises at the higher SRT, and nitrification efficiency improves to approximately 80%. Outlet ammonia returns to 4.2 mg/L, maintaining compliance through the winter period.

Model 10: Denitrification Rate Prediction

Purpose in simpler terms
This model predicts how much nitrate (produced during nitrification) will be removed by denitrification, a process where bacteria convert nitrate into nitrogen gas under low-oxygen (anoxic) conditions. Denitrification is essential for achieving the Total Nitrogen limit of <15 mg/L. In IFAS systems, much of this denitrification happens inside the biofilm, in anoxic zones where oxygen does not penetrate.

What problem this solves
- Total Nitrogen compliance depends on both nitrification (converting ammonia to nitrate) and denitrification (removing nitrate). Without predicting denitrification, the plant cannot reliably control Total Nitrogen.
- In IFAS systems, "simultaneous nitrification-denitrification" (SND) occurs naturally within the biofilm: the outer biofilm layer is aerobic (nitrification), and the inner layer is anoxic (denitrification). However, this process is not visible or directly measured, making it difficult to optimise.
- Operators do not currently have tools to estimate how much denitrification is occurring or how to adjust conditions (DO, biofilm thickness, organic carbon availability) to improve nitrogen removal.

What inputs this model uses
- **Nitrate production rate** from Model 9 (the "input" to denitrification is the nitrate produced by nitrification).
- **Dissolved oxygen levels and gradients** across the aeration tanks and within the biofilm structure. Denitrification can only occur where DO is very low (typically <0.5 mg/L).
- **Biofilm thickness estimate** from Model 11. Thicker biofilm has larger internal anoxic zones where denitrification can occur, but if too thick, the inner zones may become completely anaerobic and inactive.
- **Organic carbon availability (BOD/COD)**: Denitrifying bacteria need a carbon source (organic matter) to fuel the reaction. If BOD is very low, denitrification is carbon-limited.
- **Temperature**: Like nitrification, denitrification is also temperature-dependent, though less severely.
- **Presence of anoxic zones** downstream of aeration (if any exist in the plant design) or within the biofilm matrix.

How the model works
- The model maps the oxygen profile within the IFAS biofilm:
- The **outer layer (0–1.5 mm from the surface)** is fully aerobic (DO >1 mg/L). Nitrification occurs here; ammonia is converted to nitrate.

- The **middle layer (1.5–3 mm depth)** is a transition zone with very low oxygen (DO <0.5 mg/L). Some denitrification begins here.
- The **inner core (3–5 mm depth)** is anoxic (essentially zero DO). This is where most denitrification occurs; nitrate is converted to nitrogen gas.
- By estimating biofilm thickness and oxygen penetration depth (based on DO set-point, oxygen consumption rate, and biofilm density), the model calculates the volume of anoxic biofilm where denitrification can occur.
- It then estimates the denitrification rate based on:
- Size of anoxic zones (larger anoxic volume = more denitrification capacity).
- Nitrate availability (how much nitrate is reaching these zones from the aerobic outer layer).
- Carbon availability (whether there is enough organic matter to support denitrification).
- Temperature (denitrification slows in cold water, though not as dramatically as nitrification).
- The model predicts:
- **Denitrification rate** in mg NO_3-N removed per hour.
- **Total Nitrogen removal efficiency**: how much of the influent nitrogen is removed by the combined nitrification-denitrification process.
- **Expected effluent Total Nitrogen** concentration.
- If predicted TN is approaching the 15 mg/L regulatory limit, the model raises an alert.

Outputs and integration with other models
- Main outputs:
- **Predicted denitrification rate** (mg NO_3-N removed per hour or per day).
- **Estimated anoxic zone health and activity** within the biofilm.
- **Total Nitrogen removal efficiency** (for example, “65% of influent nitrogen is being removed; 35% remains in effluent”).
- **Forecast effluent Total Nitrogen** concentration for the next 8–12 hours.
- Integration:
- Feeds into **Model 5 (Effluent Quality Prediction)** for Total Nitrogen forecasting.
- Receives nitrification rate from **Model 9** (nitrate production is the starting point for denitrification).
- Receives biofilm thickness and health from **Model 11** (determines size of anoxic zones).
- Receives DO set-point from **Model 1**: Higher bulk DO can shrink anoxic zones in biofilm; lower bulk DO can expand them (but risks under-aerating the system).
- Informs **Model 13 (MLSS-SRT-F/M optimisation)** about nitrogen removal capacity, which can influence decisions about biomass concentration and sludge age.

Real-world example
The plant is operating with a DO set-point of 2.6 mg/L in the bulk mixed liquor. Model 11 estimates that the IFAS biofilm is approximately 3.5 mm thick on average. Model 9 predicts that

the outer 1.5 mm of biofilm is nitrifying ammonia and producing nitrate.
Model 10 analyses the oxygen gradient and estimates that the inner 2.0 mm of the biofilm is anoxic. In this anoxic zone, denitrifying bacteria are actively converting the nitrate (produced in the outer layer) back to nitrogen gas. Based on temperature (18°C), available organic carbon (moderate BOD), and the size of the anoxic zone, the model predicts that approximately 45% of the nitrate produced in the outer biofilm layer will be denitrified before it exits to the bulk liquid.
The remaining 55% of the nitrate flows out to the effluent. Combined with the influent nitrogen load, the model forecasts an effluent Total Nitrogen concentration of approximately 8.8 mg/L—comfortably below the 10 mg/L limit.
If conditions change (for example, DO set-point is raised to 3.0 mg/L to handle a shock load), Model 10 recalculates and warns that the anoxic zone in the biofilm will shrink, denitrification will decrease to approximately 30%, and effluent TN may rise to 9.5–10 mg/L, approaching the compliance limit. Operators can then decide whether the DO increase is necessary or whether other adjustments should be made to maintain nitrogen removal.

Model 11: Biofilm Media Health Monitoring

Purpose in simpler terms
This model continuously assesses the condition of the IFAS biofilm growing on the media cages. It monitors whether the biofilm is healthy, active, appropriately thick, or showing signs of stress, fouling, or degradation. Since the biofilm is a major part of the plant's nitrogen removal capacity and represents a significant capital investment, protecting and optimizing biofilm health is critical.

What problem this solves
- Biofilm condition cannot be seen directly. It is underwater, distributed across thousands of media elements, and changes gradually over weeks and months.
- Problems like biofilm fouling (where inert solids or grease coat the media and reduce performance), excessive thickening (leading to sloughing risk), or biofilm loss (due to stress or washout) are typically only noticed after treatment performance has already declined.
- There is no systematic way to track biofilm health trends or to predict when maintenance (such as media cleaning) will be needed.
- The plant invested heavily in IFAS media to improve nitrification and nitrogen removal. Without proper monitoring and care, this investment may not deliver its full potential.

What inputs this model uses
- **Oxygen Transfer Efficiency (OTE)** calculated from air flow, blower power, and measured DO.

OTE indicates how efficiently oxygen from the air is dissolving into the water. Changes in OTE over time can indicate changes in biofilm thickness or media fouling.
- **DO consumption patterns around the IFAS media zones** (“mass around cages”). High oxygen consumption near the cages indicates active, possibly thickening biofilm. Lower consumption may indicate thin or stressed biofilm.
- **Ammonia removal performance trends** from Model 9. Biofilm contributes significantly to nitrification in IFAS systems. Stable ammonia removal indicates healthy biofilm; sudden declines may indicate biofilm stress or sloughing.
- **Temperature** and **MLSS**: These affect biofilm metabolism and activity. Cold weather slows biofilm activity; high MLSS can sometimes indicate biofilm shedding and mixing with suspended solids.
- **Time since media installation or last cleaning**: Biofilm matures over weeks to months after installation or maintenance. This “biofilm age” affects performance expectations.

How the model works
- The model establishes a baseline for key indicators when the biofilm is known to be healthy (for example, shortly after installation or after cleaning):
- Oxygen Transfer Efficiency (OTE) at baseline (for example, 24–26% for new or clean media).
- Typical oxygen consumption rate around the cages.
- Stable, high nitrification efficiency.
- It then tracks changes over weeks and months:
- **OTE declining gradually** (for example, from 25% to 22% over 6 months): Suggests biofilm thickening, media surface area being partially blocked by biofilm or fouling, or accumulation of inert material.
- **Sharp increase in oxygen consumption “mass around cages”**: Indicates very active, possibly over-thickening biofilm. Risk of sloughing if thickness exceeds healthy range.
- **Sudden drop in ammonia removal without corresponding process changes**: May indicate biofilm sloughing event (biofilm detaching in large sheets, losing treatment capacity).
- **Gradual decline in ammonia removal over weeks**: May indicate biofilm stress from poor conditions (low DO, toxic influent, nutrient deficiency) or aging and fouling of media.
- Based on these patterns, the model assigns a **biofilm health score** (0–100%) and categorizes the biofilm condition:
- **Healthy (80–100%)**: Biofilm well-established, active, appropriate thickness, good nitrification performance.
- **Moderate concern (60–80%)**: Biofilm showing signs of thickening, fouling, or slight stress. Monitor closely; maintenance may be needed within 1–2 months.
- **Poor health (<60%)**: Biofilm significantly degraded, fouled, or recently lost. Immediate action required (media cleaning, process adjustment, or investigation of root cause).

- The model also estimates whether the biofilm is in a safe thickness range or approaching risky conditions (too thin, optimal, or too thick approaching sloughing risk).

Outputs and integration with other models
- Main outputs:
- **Biofilm health score** (0–100%) indicating overall condition.
- **Estimated thickness category** (light biofilm, moderate, heavy, or excessive).
- **Fouling risk indicator**: Is media performance declining due to inert material accumulation?
- **Maintenance recommendations**: For example, “Media showing signs of moderate fouling; recommend inspection and cleaning within 3–4 weeks.”
- Integration:
- Feeds into **Model 1 (DO Set-point Optimisation)**: If biofilm is stressed or too thick, Model 1 may adjust DO strategy to protect the biofilm.
- Feeds into **Model 9 (Nitrification Prediction)**: Biofilm health directly affects nitrification capacity, so Model 9 adjusts its predictions accordingly.
- Feeds into **Model 10 (Denitrification Prediction)**: Biofilm thickness affects the size of anoxic zones, which affects denitrification.
- Feeds into **Model 12 (Biofilm Stress & Washout Risk)**: Helps determine whether biofilm is in a vulnerable state.
- Receives DO control actions from **Model 1** and aeration patterns to understand operating environment for the biofilm.

Real-world example

Six months after IFAS media installation, Model 11 observes that Oxygen Transfer Efficiency has gradually declined from an initial 25% to approximately 21%. Ammonia removal performance is still good, but oxygen consumption around the media cages has increased noticeably.
The model interprets this pattern as: “Biofilm has matured and thickened over the past six months. It is now approaching the upper range of healthy thickness. Some fouling may also be present (inert solids or grease accumulating on media surfaces).”
Biofilm health score: 72% (Moderate concern).
The model issues a maintenance recommendation: “IFAS media showing signs of moderate fouling and thickening. Recommend inspection during next scheduled low-flow period. If visual inspection confirms fouling, consider media cleaning to restore oxygen transfer efficiency and prevent sloughing risk.”
Maintenance is scheduled for an upcoming weekend. During inspection, operators confirm that some media cages have visible grease and inert material accumulation. Media is cleaned (typically using high-pressure water spray or mechanical agitation), and after cleaning, OTE returns to approximately 24%. Biofilm health score improves to 88%, and the plant continues operating with strong nitrification performance.

Model 12: Biofilm Stress & Washout Risk Detection

Purpose in simpler terms
This model provides early warning of conditions that could cause sudden biofilm detachment (sloughing or “washout”), which would result in rapid loss of nitrification capacity, high effluent ammonia, and increased suspended solids. It monitors stress indicators and alerts operators 2–4 hours before a washout event is likely, giving time to take protective action.

What problem this solves
- Biofilm sloughing events are catastrophic for treatment performance. When large sections of biofilm suddenly detach, ammonia removal drops sharply, Total Nitrogen violations occur, and TSS increases in the effluent. Recovery takes days to weeks.
- These events are currently detected only after they happen (operators notice high ammonia and TSS). By that time, the damage is done.
- The root causes—such as rapid DO swings, temperature shocks, excessive biofilm thickness, or aggressive aeration cycling from ABAC (Ammonia-Based Aeration Control) systems—are not systematically monitored.
- There is no predictive tool to warn operators that the biofilm is under stress and approaching failure.

What inputs this model uses
- **Rate of change of DO (dDO/dt)**: Rapid DO swings (for example, from 1.8 mg/L to 3.2 mg/L and back down within 20–30 minutes) stress the biofilm because conditions change faster than biofilm bacteria can adapt.
- **Rate of change of MLSS (dMLSS/dt)**: A sudden increase in suspended solids can indicate biofilm detaching and mixing into the bulk liquid.
- **Aeration control patterns, especially from ABAC systems**: Ammonia-Based Aeration Control (ABAC) adjusts blowers rapidly in response to ammonia measurements. If not properly tuned, ABAC can cause rapid on/off cycling of aeration, creating DO swings that stress biofilm.
- **Temperature shock events**: Sudden cold water inflow (for example, temperature dropping >4°C in less than 2 hours due to a rain event or industrial discharge) can shock biofilm bacteria.
- **Effluent ammonia and TSS trends**: Ammonia rising despite adequate DO, combined with TSS increasing, is a classic sign of biofilm sloughing in progress.
- **Biofilm health and thickness status from Model 11**: Biofilm that is already at the upper limit of healthy thickness is more vulnerable to sloughing.

How the model works
- The model continuously monitors for stress indicators that increase the risk of biofilm

washout:
- **DO cycling rapidly** (>1.5 mg/L swing in <30 minutes): Biofilm bacteria, especially those deep in the biofilm, cannot adjust to such rapid changes. Repeated cycling weakens biofilm structure.
- **ABAC rapid on/off cycling**: If the ABAC system is aggressively turning blowers on and off every few minutes (due to ammonia fluctuations or poor tuning), this creates severe stress on biofilm.
- **Temperature shock** (drop >4°C in <2 hours): Sudden cold stress can cause biofilm bacteria to release from the media.
- **MLSS suddenly increasing + ammonia spiking simultaneously**: Strong indicator that biofilm is actively detaching and losing nitrification capacity.
- **Biofilm already at maximum safe thickness** (from Model 11): Thick biofilm has a starved inner layer that is weakly attached and prone to sudden release.
- Based on these indicators, the model calculates a **washout risk score** (0–100%):
- 0–30%: Low risk, normal operation.
- 30–60%: Moderate risk, some stress present, monitor closely.
- 60–100%: High risk, washout likely within 2–4 hours if conditions do not improve.
- When risk is high, the model generates urgent alerts and recommends immediate protective actions (described below).

Outputs and integration with other models
- Main outputs:
- **Washout risk score** (0–100%) indicating likelihood of imminent biofilm sloughing.
- **Stress indicators summary**: Which factors are contributing to risk (DO swings, ABAC cycling, temperature shock, etc.).
- **Estimated recovery timeline** if washout occurs (typically 7–14 days for biofilm to re-establish).
- **Preventive action recommendations**: Steps to reduce stress and prevent washout.
- Integration:
- Feeds into **Model 7 (Operator Advisory)** with high-priority alerts when risk is elevated.
- Coordinates with **Model 1 (DO Set-point Optimisation)**: If biofilm stress is detected, Model 1 may override aggressive ABAC cycling and stabilise DO at a steady, safe level.
- Receives biofilm health and thickness status from **Model 11**: Biofilm already at risk due to excessive thickness is more vulnerable.
- Receives temperature and anomaly data from **Model 8 (Anomaly Detection)**: Sudden temperature shocks trigger increased monitoring by Model 12.
- Can trigger automatic protective responses: For example, override ABAC control temporarily and hold DO at a stable set-point; reduce aeration intensity to minimise turbulence.

Real-world example
On a winter morning, Model 12 detects the following pattern developing:
- DO in the aeration tanks is cycling rapidly: 1.9 → 3.1 → 2.2 → 2.9 mg/L over a 25-minute

period. This is being caused by an ABAC system that is aggressively turning blowers on and off in response to ammonia fluctuations.
- Overnight, wastewater temperature dropped from 18°C to 14°C (a 4°C drop in about 6 hours—within the "cold shock" threshold).
- Model 11 had previously reported that biofilm thickness is in the upper healthy range (approaching 5 mm).
Model 12 evaluates these factors and calculates: **Washout risk = 72% (High risk zone)**.
An urgent red alert is displayed to operators: "HIGH BIOFILM STRESS – Washout risk 72% in next 2–4 hours. Immediate action recommended."
Recommended actions:
- "Override ABAC control temporarily. Stabilise DO at 2.7 mg/L using fixed set-point control."
- "Reduce blower cycling frequency. Avoid rapid on/off changes."
- "Monitor effluent ammonia and TSS closely over next 4 hours."
Operators accept the recommendations. The ABAC system is temporarily disabled, and DO is held stable at 2.7 mg/L using Model 1's standard control. Blower speed changes are made gradually rather than in rapid steps. Over the next few hours, biofilm stress subsides, the washout risk score drops to 35%, and no sloughing event occurs. Nitrification performance remains stable, and compliance is maintained.

Model 13: Coupled MLSS-SRT-F/M Optimisation

Purpose in simpler terms
This model manages the complex inter-relationships between three critical biological treatment parameters: MLSS (how much biomass is in the aeration tanks), SRT (how long biomass stays in the system, also called sludge age), and F/M ratio (the balance between incoming food/pollution and the amount of microorganisms available to eat it). These three parameters are tightly linked—changing one affects the others—and must be optimised together, not separately.

What problem this solves
- Operators often adjust MLSS, SRT, or F/M individually without fully considering the knock-on effects. For example:
- Increasing MLSS (to handle higher load) without adjusting SRT can create an impossible biological condition.
- Extending SRT (to improve nitrification) without considering the resulting change in F/M ratio can lead to bulking or energy waste.
- These three parameters cannot be independently controlled—they are mathematically and biologically coupled. Trying to control them separately leads to conflicts and instability.

- There is also a tradeoff between treatment performance and energy consumption: Higher MLSS provides more treatment capacity but requires more oxygen (higher energy); lower MLSS uses less energy but has less treatment capacity and stability.

What inputs this model uses
- **Current MLSS** measured by turbidity sensors or lab tests (typically target range 3,000–4,500 mg/L for this plant).
- **Current SRT** calculated from the waste activated sludge (WAS) flow rate and the total biomass inventory (SRT = Total biomass ÷ Biomass wasted per day).
- **Influent BOD/COD load** (determines the "Food" in the F/M ratio).
- **Current F/M ratio** calculated as: (kg BOD per day) ÷ (kg MLSS in aeration tanks).
- **Oxygen consumption and energy use** from blowers (from Model 3): Higher MLSS means more oxygen demand.
- **Effluent quality requirements** from Model 5: Whether there is margin to reduce MLSS or whether higher MLSS is needed to maintain compliance.

Effluent quality requirements: BOD <10 mg/L, TSS <10 mg/L, TN <10 mg/L (NGT 2019 metropolitan standards)

- **Temperature** (from seasonal monitoring): Cold weather requires higher SRT for nitrification, which affects MLSS and F/M balance.

How the model works
- The model recognises the fundamental interdependencies:
- **MLSS ↔ SRT**: If you increase SRT (by reducing WAS flow), biomass accumulates and MLSS rises. If you reduce SRT (increase WAS), MLSS falls.
- **SRT ↔ F/M**: If influent load is constant but SRT increases (and therefore MLSS increases), F/M ratio decreases (more microorganisms per unit of food).
- **F/M ↔ Treatment performance**: Very low F/M (<0.10) can lead to starvation conditions, favouring filamentous bacteria and causing bulking. Very high F/M (>0.40) can overload the system and cause poor effluent quality.
- **MLSS ↔ Energy**: Higher MLSS requires more oxygen supply because the larger biomass population consumes more oxygen just for respiration.
- The model solves a multi-objective optimisation problem:
- **Maximise treatment quality**: BOD, TSS, ammonia, and TN all within regulatory and internal targets.
- **Minimise energy consumption**: Run with the minimum MLSS and oxygen supply that still guarantees compliance.
- **Maintain biological stability**: Keep F/M and SRT within ranges that prevent bulking and

support stable nitrification.
- **Subject to constraints**: MLSS must stay within 3,000–4,500 mg/L; SRT must stay within 12–25 days (IFAS optimal range); F/M should remain in 0.15–0.30 range (healthy activated sludge conditions).
- The model calculates optimal values for all three parameters together:
- Target MLSS concentration.
- Target SRT.
- Resulting F/M ratio (checked to ensure it is within healthy range).
- Required adjustments: WAS flow rate (controls SRT), RAS flow rate (helps adjust MLSS), DO set-point from Model 1 (supports the biomass and treatment).
- It prevents operators or other models from making contradictory changes. For example, if an operator tries to increase MLSS without correspondingly reducing WAS (increasing SRT), the model will flag this as inconsistent and recommend the full set of coordinated changes needed.

Outputs and integration with other models
- Main outputs:
- **Recommended MLSS set-point** (for example, “Maintain MLSS at 3,800 mg/L for current conditions”).
- **Recommended SRT target** (for example, “SRT = 18 days; reduce WAS flow to 38 m^3/hr to achieve this”).
- **Calculated F/M ratio** and confirmation that it is within healthy range.
- **Coordinated WAS and RAS flow set-points** to achieve the recommended MLSS and SRT.
- Integration:
- Coordinates closely with **Model 3 (Energy Optimisation)**: The MLSS and oxygen demand targets from Model 13 are used by Model 3 to optimise blower and mixer energy.
- Coordinates with **Model 14 (Seasonal SRT Adjustment)**: Model 14 provides temperature-driven SRT targets; Model 13 ensures that MLSS and F/M are adjusted accordingly.
- Feeds into **Model 5 (Effluent Quality Prediction)**: Changes in MLSS, SRT, and F/M affect predicted treatment performance.
- Feeds into **Model 15 (Bulking Prevention)**: F/M ratio is a critical indicator for bulking risk; Model 13 ensures F/M stays in safe range.
- Receives nitrification requirements from **Model 9**: If nitrification is struggling, Model 13 may increase SRT to give nitrifiers more time.

Real-world example
Influent BOD increases suddenly from an average of 250 mg/L to 380 mg/L due to an industrial discharge event. Operators are considering increasing MLSS to handle the higher load.
Without Model 13 (manual operation):
- Operator increases RAS flow to raise MLSS from 3,500 to 4,200 mg/L.

- But WAS flow is left unchanged, so SRT does not change significantly.
- F/M ratio rises from 0.22 to 0.32 (higher load, more MLSS, but not enough SRT adjustment).
- Oxygen demand spikes (more biomass consuming oxygen). Energy increases sharply.
- F/M is now at the upper edge of the safe range; there is some risk of treatment quality declining if the high load continues.

With Model 13 (coordinated optimisation):
- Model 13 detects the BOD spike.
- It calculates: "Optimal response for this higher load: Maintain MLSS at 3,900 mg/L (moderate increase), Reduce WAS slightly to increase SRT to 18 days (from current 16 days), Raise DO set-point to 2.7 mg/L (from current 2.5 mg/L)."
- F/M ratio remains at 0.24 (within healthy range). SRT is extended to support nitrification under slightly higher load. DO is increased to provide adequate oxygen.
- All three parameters (MLSS, SRT, F/M) are adjusted together in a coordinated way.
- Result: Treatment quality remains stable (BOD and ammonia within targets). Energy increase is minimised (MLSS increase is moderate, and DO increase is only what is needed). Biological stability is maintained (F/M in safe range, SRT appropriate).

Model 14: Temperature-Driven Seasonal SRT Adjustment

Purpose in simpler terms

This model automatically adjusts sludge age (SRT) based on wastewater temperature to compensate for the slower growth of nitrifying bacteria in cold weather. By proactively extending SRT as winter approaches and shortening it again in summer, the model prevents the predictable nitrification failures that occur every winter in manually operated plants.

What problem this solves

- Nitrifying bacteria are extremely sensitive to temperature. In winter (10–15°C), they grow 3–5 times more slowly than in summer (25–30°C).
- With fixed SRT operation, winter cold causes nitrifier populations to decline or wash out, leading to ammonia and Total Nitrogen violations every year from December through February.
- Operators know this happens, but without clear guidance, they adjust SRT too late, too little, or inconsistently.
- The IFAS biofilm provides some cold-weather stability, but only if SRT is managed properly to allow nitrifiers time to build stable populations in both the suspended phase and the biofilm.

What inputs this model uses

- **Current wastewater temperature** measured continuously at the inlet.
- **Temperature forecast trends** from weather data or historical seasonal patterns.

- **Historical temperature-nitrification performance correlations**: Data showing how nitrification efficiency changes with temperature at this specific plant.
- **Current nitrification efficiency** from Model 9 (actual performance feedback).
- **Biofilm nitrifier population status** from Model 11 (biofilm bacteria are somewhat more cold-tolerant).

How the model works
- The model tracks wastewater temperature continuously and identifies seasonal transitions (summer → autumn → winter → spring → summer).
- It applies temperature-based adjustment rules for SRT derived from established biological principles and calibrated to the plant's specific performance:
- **>25°C (Summer):** Nitrifiers grow quickly; SRT = 12–15 days is sufficient.
- **20–25°C (Mild/Transition):** Nitrifiers grow at moderate rate; SRT = 15–18 days (baseline).
- **15–20°C (Cool autumn or spring):** Nitrifier growth slowing; SRT = 18–20 days (+15–20% increase).
- **10–15°C (Winter cold):** Nitrifier growth severely slowed; SRT = 22–30 days (+50–100% increase compared to summer).
- **<10°C (Extreme cold, rare in Ahmedabad but possible in other locations):** SRT = 30+ days; maximum extension needed.
- The model makes adjustments gradually over days or weeks as temperature changes, rather than sudden large changes. For example, as autumn temperatures decline from 22°C to 16°C over a three-week period, SRT is gradually increased from 16 days to 19 days in several small steps.
- The adjustment is coordinated with Model 13 to ensure that MLSS and F/M ratio remain in acceptable ranges as SRT changes.
- The model also coordinates with Model 1 (DO set-point) because winter operation often requires both higher SRT (longer retention time for nitrifiers) and slightly higher DO (to compensate for slower biological activity).

Outputs and integration with other models
- Main outputs:
- **Temperature-adjusted SRT target** for current season (for example, "Current temperature 14°C; recommended SRT = 23 days").
- **WAS flow rate adjustment** to achieve the target SRT (reducing WAS flow extends SRT; increasing WAS flow shortens SRT).
- **Seasonal alerts and transition warnings**: For example, "Winter approaching; prepare for SRT extension over next 2 weeks" or "Spring warming detected; begin gradual SRT reduction."
- Integration:
- Coordinates closely with **Model 13 (MLSS-SRT-F/M)**: Model 14 provides the

temperature-driven SRT target; Model 13 ensures MLSS and F/M are adjusted accordingly.
- Feeds into **Model 9 (Nitrification Prediction)**: The adjusted SRT directly affects nitrifier population and nitrification capacity.
- Coordinates with **Model 1 (DO Set-point)**: Winter operation typically requires both higher SRT and slightly elevated DO.
- Feeds into **Model 2 (Aeration Demand Forecasting)**: Seasonal SRT changes affect overall oxygen demand and aeration requirements.
- Receives feedback from **Model 9**: If nitrification efficiency is still declining despite SRT increase, Model 14 may recommend further extension.

Real-world example
In mid-November, wastewater temperature begins declining steadily from 22°C. Model 14 detects the trend and predicts that by early December, temperature will be in the 14–16°C range.
The model issues a proactive advisory: "Seasonal transition to winter conditions detected. Temperature declining from 22°C to expected 15°C over next 3 weeks. Nitrification efficiency will decline by approximately 35–40% without adjustment."
Recommended action plan:
- Week 1 (current temp 22°C → 20°C): Increase SRT from current 15 days to 18 days (reduce WAS flow by 15%).
- Week 2 (predicted temp 18°C): Increase SRT to 20 days (reduce WAS further by 10%).
- Week 3 (predicted temp 15–16°C): Increase SRT to 22 days (final winter target).
- Coordinate with Model 1: Gradually raise DO set-point from 2.5 mg/L to 2.7 mg/L to support nitrification at lower temperature.
These adjustments are implemented gradually and automatically (or with operator approval). By the time full winter cold arrives in December, the plant is already operating at winter SRT and DO settings. Nitrification efficiency is maintained at approximately 78–82% (compared to 90%+ in summer, but sufficient to maintain compliance). Effluent ammonia remains below 5 mg/L and Total Nitrogen remains below 9 mg/L throughout the winter. No violations occur. In March, as temperature begins rising, Model 14 detects the warming trend and gradually reduces SRT and DO back to baseline settings, maintaining optimal operation year-round.

Temperature Range	SRT Adjustment	WAS Flow Change	Nitrification Risk	Operator Alert
>25°C (Summer)	-10% (12-15 days)	+10% more WAS	LOW	None

Temperature Range	SRT Adjustment	WAS Flow Change	Nitrification Risk	Operator Alert
20-25°C (Mild)	Baseline (15 days)	Baseline	LOW	None
15-20°C (Cool)	+20% (18-20 days)	-20% less WAS	MEDIUM	Monitor TN
10-15°C (Winter)	+50% (22-30 days)	-50% less WAS	HIGH	Prepare winter mode
<10°C (Extreme Cold)	+100% (30+ days)	-60% WAS flow	CRITICAL	Alert operator

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TIER 3: ADVANCED Optimisation MODULES (Models 15-20)

Specialised predictive and optimisation models for sludge bulking prevention, digester performance, nutrient balance, and final effluent quality assurance.

Model 15: Sludge Bulking & Filamentous Growth Early Warning

Purpose in simpler terms
This model continuously looks for early signs that the sludge in the aeration and clarifier system is moving toward "bulking" conditions, where it stops settling properly and starts floating. It predicts this risk several days in advance so that corrective actions can be taken before clarifier performance and effluent quality are affected.

What problem this solves
- Bulking sludge is usually discovered only when clarifier blankets rise, effluent TSS increases, and visible solids escape over the weirs. At that point, recovery typically takes 1–2 weeks.
- The main drivers of bulking (very low F/M ratio, very long SRT, low DO, certain temperature

and load combinations) usually develop gradually over days. Without a predictive tool, this slow drift goes unnoticed.
- Bulking leads to solids carryover, TSS violations, and can indirectly cause BOD and nitrogen problems by reducing effective clarification capacity.

What inputs this model uses
- **F/M ratio trends** from Model 13 (low food-to-microorganism ratios, especially when <0.15 for extended periods, favour filamentous organisms).
- **SRT trends** (very high sludge age >25 days combined with low F/M and low DO can promote bulking).
- **DO levels and stability** in aeration (sustained low DO or large DO swings are risk factors).
- **Temperature**: certain temperature bands, combined with operating conditions, can increase filament growth potential.
- **Historical SVI (Sludge Volume Index)** or settling test results, if available, to train the model on how bulking manifested in the past.
- **Recent operational changes** such as large MLSS increases, major load shifts, or changes in recycle flows.

How the model works
- It learns from past plant data what the typical "lead-up pattern" to bulking looks like:
- Example: F/M gradually falling below 0.15, SRT above 25 days, DO slightly low (1.5–2.0 mg/L), followed by SVI increasing from 100 to 130 to 160 mL/g over 5–7 days.
- It continuously calculates a **bulking risk score** based on how closely current conditions match these risky patterns.
- It also projects SVI trends (if historical SVI data are available) or equivalent settling performance indicators, estimating where these will be in 3–7 days if no changes are made.
- When the risk crosses certain thresholds, it classifies the situation as:
- Low risk: monitor, no action required.
- Medium risk: recommend preventive adjustments (e.g. slightly raise F/M by reducing MLSS or increasing load if available, raise DO).
- High risk: strong recommendation to implement corrective actions immediately, before clarifier performance visibly deteriorates.

Outputs and integration with other models
- Main outputs:
- **Bulking risk score** (for example, 0–100%).
- **Predicted short-term settling performance** (SVI or equivalent indicators 3–7 days ahead).
- **Root-cause oriented recommendations**, such as "F/M too low for extended time; reduce MLSS or adjust loading", or "DO too low; increase aeration in specific tanks."

- Integration:
- Works with **Model 13 (MLSS-SRT-F/M)** to recommend changes in MLSS, SRT, and F/M in a coordinated way, not in isolation.
- Feeds into **Model 7 (Operator Advisory)** to raise early alerts and provide practical operator guidance.
- Uses DO control information from **Model 1** to identify whether poor oxygen control is contributing to bulking risk.
- Interacts with **Model 4** because bulking often increases polymer demand; early prevention reduces downstream chemical usage.

Real-world example
Over a two-week period, the model observes that F/M has fallen from 0.22 to 0.13 while SRT has gradually risen from 18 to 26 days. DO in the aeration tanks is stable but on the low side (1.6–1.8 mg/L). Historical patterns show that similar combinations previously led to SVI rising above 140–150 mL/g within a week.
Model 15 calculates a bulking risk score of 65% and predicts that, if nothing changes, settling performance will deteriorate significantly in about 5 days.
It recommends:
- "Reduce MLSS by approximately 15% over the next few days (slightly increase WAS flow) to bring F/M back into the 0.18–0.22 range."
- "Increase DO to around 2.2 mg/L in critical aeration zones to discourage filamentous growth."
Operators implement these measures. As a result, F/M returns to 0.20, SRT drops to around 20 days, and the bulking risk score falls back below 30%. Clarifier performance remains stable, and a potential bulking crisis is avoided.

Model 16: Filamentous Organism Identification (Advanced)

Purpose in simpler terms
This model helps differentiate between types of filamentous microorganisms likely to be causing bulking, based on process conditions and, where available, microscopic or image data. Different filament species prefer different conditions, so identifying the likely type supports more targeted corrective strategies.

What problem this solves
- When bulking occurs, the generic response is often "raise DO and maybe dose some chemicals," but different filament types respond to different interventions.
- Without understanding which organisms are likely dominant (for example, Microthrix, Nocardia, or others), some standard responses may be ineffective or even make the situation

worse.
- Microscopic examination is powerful but not always interpreted consistently. Many plants do not have on-site microbiology expertise.

What inputs this model uses
- **Process condition patterns**: sustained F/M, SRT, DO levels, presence of fats/oils/grease, pH, temperature. Different organisms favour distinct combinations.
- **Historical correlations**: records linking lab/ microscopy identifications with specific upstream and process conditions (if available).
- **Optional microscopic or image data**: if the plant provides microscope images, the model can use pattern recognition to support identification (e.g. filament length, branching, location in floc).
- **Bulking risk and context** from Model 15.

How the model works
- It maps current operating conditions and upstream influent characteristics against known "profiles" of common filamentous organisms. For example:
- **Microthrix** is often associated with low F/M, long SRT, presence of fats/oils/grease, and cooler temperatures.
- **Nocardia** is frequently linked to high grease and scum formation.
- Other filaments may correlate with low DO, specific pH ranges, or certain industrial inputs.
- If microscopy or image recognition is available, the model refines its assessment by matching observed shapes and locations of filaments in the flocs.
- It then outputs a likely dominant organism or small set of candidates, each with a confidence rating, and suggests tailored corrective actions (e.g. adjust F/M, remove grease sources, increase DO, adjust SRT).

Outputs and integration with other models
- Main outputs:
- **Likely filament type(s)** (e.g. "Conditions are most consistent with Microthrix dominance – high confidence")
- **Targeted control recommendations** matched to the suspected organism.
- Integration:
- Works hand-in-hand with **Model 15 (Bulking Early Warning)** to explain "what kind" of bulking is likely and how best to respond.
- Feeds into **Model 7 (Operator Advisory)** for easy-to-understand microbiology guidance without requiring deep microbiology expertise.
- Helps validate or refine operator and lab observations, improving long-term process knowledge.

Real-world example
The plant begins experiencing foaming and a slow increase in sludge blanket height. Model 15 flags an elevated bulking risk, while Model 16 analyses the conditions: low F/M (~0.13), high SRT (~28 days), significant fats/oils/grease in influent from nearby food industries, and moderate DO.
The model concludes that **Microthrix parvicella** is the most likely dominant filament type and recommends:
- "Increase F/M slightly by reducing MLSS or adjusting loading where possible."
- "Work with upstream sources to improve grease trapping and reduce fats/oils/grease entering the plant."
- "Avoid aggressive chlorination of the aeration tanks as a first response, as it may not effectively resolve Microthrix and can harm floc-forming bacteria."
Operators focus on upstream grease control and carefully adjust MLSS and F/M. Over time, settling improves and foaming reduces, confirming that the targeted response was appropriate.

Model 17: Anaerobic Digester Biogas Production Optimisation

Purpose in simpler terms
This model optimises how the anaerobic digester is fed and mixed so that it produces as much stable biogas as possible from the sludge, while maintaining digester health. It treats the digester as a biological reactor that can be tuned rather than just a fixed "black box."

What problem this solves
- With fixed feed and mixing, the digester may run below its potential, producing less biogas than it could from the same amount of sludge.
- Overfeeding, underfeeding, or inappropriate mixing can cause digester instability, foaming, or acidification, which then take many days to correct.
- There is often no real-time link between what is being fed to the digester (in terms of solids and energy content) and the biogas produced.

What inputs this model uses
- **WAS (waste activated sludge) feed rate** to the digester: volume and total solids concentration.
- **Volatile solids content** of the sludge (the biodegradable portion that can be converted to biogas).
- **Digester temperature** (especially important for mesophilic digestion around 35–37°C).
- **Mixing pump speed and operation duty cycle** in the digester.
- **Real-time biogas production rate** and, if available, methane content.
- **Indicators of digester health**: pH, alkalinity, and volatile fatty acids (VFAs), if measured.

How the model works

- It estimates the **organic loading rate** to the digester (kg of volatile solids per cubic meter of digester volume per day) and compares this with optimal ranges derived from design and historical behaviour.
- It correlates past data between:
- Loading rate and biogas production.
- Temperature stability and biogas yield.
- Mixing intensity and gas production and stability.
- By continuously learning from how changes in feed and mixing affect gas output and stability, the model identifies feed and mixing patterns that maximise biogas production without pushing the digester into an unstable state.
- It then recommends (or directly applies) adjustments to:
- Sludge feed rate (within constraints set by sludge production and storage).
- Mixing duty cycle and intensity, ensuring there is enough mixing to avoid stratification but not so much that it wastes energy or disturbs bacterial communities.
- It also provides forecasts of expected biogas production for the next 24–48 hours based on current and planned feed rates.

Outputs and integration with other models

- Main outputs:
- **Recommended digester feed rate profile** (e.g. more evenly distributed feeding instead of large slugs).
- **Recommended mixing strategy** (for example, 70–80% duty cycle during peak gas production periods, less during off-peak).
- **Predicted biogas production curve** for planning and monitoring.
- Integration:
- Receives WAS flow and sludge characteristics from **Model 13** (which determines how much WAS is generated).
- Feeds into **Model 3 (Energy Optimisation)**, which can consider biogas energy as part of the plant's energy balance where gas is used for heat or power.
- Coordinates with **Model 18 (Digester Upset Detection)** to ensure that optimisation stays within safe operating boundaries.

Real-world example

The digester is currently operated with a fixed feed of 40 m³/day of WAS and continuous mixing at 100% speed. Biogas production is steady but not very high.

Model 17 examines historical data and finds that when feed is smoothed out over the day (instead of fed in one or two large batches) and mixing is run at about 75–80% duty cycle (for example, 15 minutes on, 5 minutes off), biogas production and stability both improve.

The model recommends:
- “Feed WAS continuously at 35–45 m^3/day, adjusted based on daily solids loading.”
- “Operate digester mixing at 80% duty cycle rather than 100% continuous.”
After these changes are implemented, biogas production increases and remains stable, and no signs of foaming or upset are observed.

Model 18: Anaerobic Digester Upset Detection & Recovery

Purpose in simpler terms
This model acts as a protection system for the anaerobic digester. It detects early signs of digester stress or upset (such as acid build-up, load shock, or temperature problems) and automatically reduces the feed or adjusts conditions so that the digester can recover before a full-scale failure occurs.

What problem this solves
- When a digester goes into upset (acidification, foam, severe gas drop), it can take many days to recover. During this time, sludge may have to be bypassed or handled less efficiently.
- Early warning signs of problems—slight drops in gas output, small pH dips, VFA increases—are often not recognised in time.
- Without automatic response, operators may continue feeding at normal rates even as the digester is struggling, making the upset worse.

What inputs this model uses
- **Biogas production rate** and trends (rapid or sustained decline is a key indicator).
- **pH** in the digester (acid build-up causes pH to fall).
- **Temperature** (digester bacteria are very sensitive to temperature shocks or deviations from the ideal range).
- **VFAs and alkalinity** if available (VFAs increase and alkalinity is consumed during upset).
- **Feed rate history** from Model 17 and sludge characteristics (sudden large increases in feed increase risk).

How the model works
- It establishes normal operating ranges and patterns for the digester when it is healthy, including normal daily fluctuations in gas production.
- It then monitors for deviations such as:
- **Biogas production dropping faster than expected** without a corresponding drop in feed rate.
- **pH decreasing** below safe values (for example, dropping from ~7.1 toward 6.8 or lower).
- **Temperature drifting away** from the target set-point or large fast changes.
- **VFA increasing and alkalinity decreasing** (if measured), classic signs of acidification.

- When such patterns are detected, the model classifies the level of concern and, at higher levels, recommends or implements automatic changes such as:
- Reducing WAS feed rate to the digester.
- Adjusting mixing (sometimes too much mixing during an upset can worsen foam or disturbance).
- Raising temperature carefully back to target if it has dropped.
- It also estimates how long the digester is likely to take to recover, based on past recoveries and the current severity.

Outputs and integration with other models
- Main outputs:
- **Digester health score** and **upset risk level** (for example, Normal, Caution, High Risk).
- **Recommended feed rate reduction** or other actions during upset conditions.
- **Estimated recovery time** under current intervention plan.
- Integration:
- Works with **Model 17** to override optimisation and move into protective mode when upset risk is high.
- Feeds into **Model 7 (Operator Advisory)** with clear warnings and action steps.
- Receives relevant data from plant instrumentation and any laboratory VFA/alkalinity measurements.

Real‑world example
The digester has been running normally. Over two days, Model 18 detects that biogas production has fallen from its usual level by about 20%, even though feed rate has not changed, and pH is slowly dropping from 7.1 to 6.9.
The model flags this as a developing upset: "Caution: Digester performance declining. Early signs of acidification likely."
It recommends automatically reducing WAS feed by 25% and closely monitoring pH and gas output, while advising operators to check recent changes in sludge characteristics or upstream processes.
Operators accept the recommendation. Over the next few days, gas production stabilises and pH stops falling, then slowly recovers. Once the digester returns to stable operation, feed is gradually increased back to normal. A full upset is avoided.

Model 19: Nutrient Balance Monitoring & Deficiency Warning

Purpose in simpler terms
This model checks whether the wastewater has the right balance of nutrients—carbon (BOD),

nitrogen (N), and phosphorus (P)—for healthy biological treatment. If key nutrients are lacking, the biology cannot function properly even if DO and SRT are controlled. The model warns when nutrient imbalance is likely to harm treatment performance.

What problem this solves
- Most biological treatment processes assume that wastewater has enough nitrogen and phosphorus relative to organic carbon. However, in some cases (for example, from industrial sources or highly diluted streams), nitrogen or phosphorus can be deficient.
- If nitrogen is too low, nitrification and biomass growth are limited, even if all other conditions seem correct.
- If phosphorus is too low, bacteria cannot synthesize the cell material they need for growth, leading to poor floc formation and unstable operation.
- These deficiencies can appear as unexplained poor performance if nutrient ratios are not explicitly checked.

What inputs this model uses
- **Daily or frequent measurements of influent BOD (or COD)**.
- **Influent Total Nitrogen (TN)** measurements or estimates.
- **Influent Total Phosphorus (TP)** measurements or estimates.
- **Plant flow** to calculate mass loads (kg/day of BOD, N, and P).

How the model works
- It calculates the ratio of BOD : N : P in the influent, which is often compared to an ideal ratio for conventional biological treatment (for example, around 100:5:1 for BOD:N:P).
- It then assesses whether nitrogen or phosphorus is below the approximate minimum needed to support:
- Biomass growth and maintenance.
- Nitrification and denitrification processes.
- If the BOD:N or BOD:P ratios are outside healthy bands for extended periods, the model predicts reduced biological performance and flags a potential nutrient deficiency.
- It may also estimate how much supplemental nutrient (such as ammonium or phosphate salts) would be required to restore a healthy balance if supplementation is considered as a strategy.

Outputs and integration with other models
- Main outputs:
- **Current BOD:N:P ratios** and simple interpretation (“balanced”, “nitrogen-deficient”, “phosphorus-deficient”).
- **Deficiency warnings** when imbalances are likely to affect biology, along with suggested mitigation approaches.
- Integration:

- Provides context to **Model 9 (Nitrification)** when nitrification performance is poorer than expected despite seemingly correct SRT and DO—indicating possible nitrogen deficiency.
- Informs **Model 13 (MLSS-SRT-F/M)** because nutrient deficiency can change how biomass responds to SRT/F/M adjustments.
- Displays clear messages via **Model 7 (Operator Advisory)** so operators can discuss possible nutrient supplementation or process changes with process engineers.

Real-world example
Over several weeks, records show that influent BOD remains around 200–250 mg/L, but Total Nitrogen has dropped significantly due to upstream process changes at major industrial contributors. The calculated BOD:N ratio drifts from a healthy 100:6 to a potentially problematic 100:3.
Model 19 identifies this as nitrogen deficiency and flags: "Nitrogen availability is below optimal level. Nitrification and biomass growth may be limited even if SRT and DO are correct."
Operators and process engineers review this information and, if appropriate for the site, decide whether to:
- Blend in other streams with higher nitrogen content, or
- Add a small amount of nitrogen supplement to restore balance.
As a result, nitrification performance stabilises, and the "mysterious" performance issues gain a clear biochemical explanation.

Model 20: Chlorine Residual Optimisation

Purpose in simpler terms
This model ensures that the disinfection step at the end of the plant is effective and efficient. It adjusts chlorine dosing so that harmful microorganisms are reliably killed and regulatory microbiological standards are met, while avoiding unnecessary excess chlorine in the final effluent.

What problem this solves
- With fixed chlorine dosing, the plant often operates at a dose intended for worst-case conditions, even when the effluent is very clean. This results in over-chlorination and higher residual chlorine in the discharge than necessary.
- If the dose is set too low for a given combination of effluent quality and contact time, disinfection may be incomplete and microbiological limits could be breached.
- There is often no dynamic link between effluent quality, hydraulic contact time, and chlorine dose.

What inputs this model uses
- **Effluent BOD, TSS, and turbidity**: Higher values shield microorganisms from chlorine and increase chlorine demand.
- **Effluent flow rate**: Determines how quickly water passes through the chlorine contact chamber and therefore how much time chlorine has to act.
- **Contact chamber volume and hydraulics**: Used to calculate effective contact time.
- **Measured chlorine residual** at the outlet of the contact chamber.
- **Microbiological standards** (for example, required fecal coliform limits and any internal safety margins).

How the model works
- It calculates the required **CT value (Concentration × Time)** for disinfection based on current effluent quality and regulatory guidelines.
- Knowing the actual hydraulic retention time in the contact chamber (calculated from flow and volume), it determines what chlorine concentration is needed to achieve the desired CT.
- It then sets the chlorine dose so that:
- The target CT is achieved for effective disinfection.
- The measured residual at the outlet remains within a specified range (for example, 0.5–1.0 mg/L).
- The model uses a feedback loop: if measured residual is consistently higher than target, it reduces dose slightly; if residual is too low or disinfection performance is at risk, it increases dose.
- Over time, it learns how different effluent quality conditions respond to chlorine addition, improving its predictions and fine-tuning.

Outputs and integration with other models
- Main outputs:
- **Optimal chlorine dose set-point** (mg/L) for the current conditions.
- **Predicted residual chlorine** at the contact chamber outlet.
- **Disinfection safety margin** relative to regulatory requirements.
- Integration:
- Receives effluent quality predictions from **Model 5** to anticipate changes and adjust chlorine dosing preemptively.
- Sends set-points to the chlorine dosing system and residual information to **Model 7 (Operator Advisory)** for operator monitoring.
- Complements **Model 4**, which addresses chlorine use from a broader chemical optimisation perspective.

Real-world example
During a period of good operation, effluent BOD is around 3 mg/L, TSS around 5 mg/L, and flow is moderate, giving a relatively long contact time in the chlorine contact chamber. Model 20 calculates that a chlorine dose of 1.2–1.3 mg/L is sufficient to achieve the required disinfection and leave a residual of about 0.6 mg/L.
On a different day, heavy rain causes effluent TSS to rise slightly and flow to increase, shortening the contact time. The model recalculates and finds that a higher dose, around 1.8–1.9 mg/L, is needed to maintain the same disinfecting CT and residual.
In both cases, the chlorine dose is matched to the real need, ensuring reliable disinfection without chronic over-chlorination.

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6. CRITICAL INTER-DEPENDENCIES & PROCESS LINKAGES

1. MLSS ↔ SRT ↔ F/M ↔ DO ↔ Nitrification Rate

The Relationship:

High F/M (fast-fed) → Faster metabolism → Higher DO demand
Low F/M (slow-fed) → Slowed metabolism → Lower DO need, but extended SRT needed for nitrification

Model Integration: - **Model 13 coordinates:** If influent BOD spikes, simultaneously lower MLSS (prevent overload accumulation), raise DO (support higher nitrification), adjust WAS (maintain target SRT) - **Model 14 seasonal:** Winter cold → need higher SRT for nitrification → Model 13 accepts this constraint - **Model 9-10 monitor:** Nitrification efficiency validates if DO/SRT adjustments working

Operational Impact: Without coordination, operator raising MLSS alone would increase F/M (good nitrification potential) but create oxygen demand exceeding blower capability → system stalls

2. Temperature ↔ Nitrification ↔ SRT ↔ Biofilm Activity

The Relationship:

Winter (10-15°C) → Nitrifier growth rate: -50% → Need SRT: +50% → Biofilm protects (slower decline than suspended)
Summer (>25°C) → Nitrifier growth rate: baseline → Need SRT: baseline → Biofilm/suspended comparable

Model Integration: - **Model 14 automatic:** Temperature drops → automatically increase SRT via WAS reduction - **Model 9 validates:** Nitrification efficiency maintained even in winter - **Model 12 watches:** Biofilm stress during temperature shock (sudden cold inflow)

Operational Impact: Without Model 14, Dec-Feb winter operation would inevitably drop nitrification efficiency to 50-60%, causing TN compliance failure (historical problem). With Model 14, efficiency maintained at 80%+

3. BOD Load ↔ Nitrification ↔ Denitrification ↔ Total Nitrogen

The Relationship:

High BOD → Nitrification needs more O2 → Less space for denitrification in biofilm → Lower TN removal
Low BOD → Efficient nitrification → More biofilm anoxic zone → Better denitrification → Lower TN
BUT: Very low BOD + low F/M → Filament growth → bulking risk → treatment failure

Model Integration: - **Model 9-10:** Predict combined nitrification + denitrification efficiency - **Model 13:** Balance F/M to maintain both nitrification AND denitrification potential - **Model 15:** Monitor filament risk if F/M drops too low during low-BOD periods

Operational Impact: Simple high-sludge-age operation reduces nitrification. Model 9-10 coupled approach with Model 13 ensures BOTH nitrification (biofilm/suspended) AND denitrification (biofilm pores) optimised simultaneously

4. Biofilm Health ↔ Oxygen Transfer ↔ Aeration Energy ↔ Treatment Efficiency

The Relationship:

Clean biofilm → High oxygen transfer (OTE) → Efficient nitrification → Low energy for similar treatment
Fouled biofilm → Low oxygen transfer → Need higher aeration/DO → Energy waste + low TN

Model Integration: - **Model 11:** Estimates biofilm condition via OTE efficiency - **Model 1-2:** Adjust DO/aeration based on biofilm OTE (fouled media need higher aeration) - **Model 3:** Energy Optimisation knows fouled media requires more blower power - **Model 12:** Detects stress events that damage biofilm

Operational Impact: Maintenance decision: If biofilm fouling detected, operator can plan media cleaning during low-flow period rather than emergency response during compliance crisis

5. WAS Rate ↔ SRT ↔ Digester Feed ↔ Biogas ↔ Energy Balance

The Relationship:

Higher WAS flow → Lower SRT → Lower biofilm age → Higher solids to digester → More biogas
Lower WAS flow → Higher SRT → Older biofilm → Lower solids to digester → Less biogas
BUT: SRT must be high enough for nitrification → energy Optimisation limited by treatment requirement

Model Integration: - **Model 13:** Sets WAS flow to achieve SRT needed for nitrification (primary constraint) - **Model 17:** Given WAS rate, predicts digester biogas production - **Model 3:** Considers biogas energy potential in overall energy balance - **Model 18:** Monitors digester health for upsets

Operational Impact: Cannot Maximise biogas without sacrificing nitrification control. Model integration ensures optimal balance within NGT compliance constraints

6. Nitrifier Population ↔ DO Stability ↔ Sludge Bulking Risk

The Relationship:

Stable DO (3-4 mg/L) → Stable nitrifier growth → Filaments suppressed → Good settling
Fluctuating DO (2-4 mg/L) → Nitrifier stress → Filament growth → Poor settling → Return sludge loss

Model Integration: - **Model 1:** Maintains stable DO within ±0.5 mg/L (vs. manual ±1.5 mg/L swings) - **Model 8:** Detects shock loads that cause DO swings - **Model 15:** Monitors filament growth risk linked to DO instability - **Model 12:** Detects biofilm stress from sudden aeration changes

Operational Impact: Manual DO control with large swings would indirectly cause bulking through nitrifier stress and filament competitive advantage. Model 1 precise control prevents this mechanism

7. Chemical Demand ↔ Coagulation ↔ Sludge Production ↔ SRT ↔ Digester Loading

The Relationship:

Excess coagulant → Overdose polymer → Improved cake → BUT excess solids to digester → Digester upset risk
Optimised chemical → Right polymer dose → Good cake + controlled SRT → Balanced digester load

Model Integration: - **Model 4:** Optimises polymer/$FeCl_3$ for actual sludge characteristics (not overdose) - **Model 3:** Energy Optimisation includes reduced chemical cost - **Model 13:** SRT maintained at target despite reduced chemical output - **Model 17-18:** Digester receives optimised solids feed

Operational Impact: Without Model 4 Optimisation, excess polymer (safety margin mentality) would increase WAS solids content, requiring more digester capacity or causing upsets during high-load periods

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7. IMPLEMENTATION STRATEGY

Phased Deployment Approach

Phase 1: Design & Foundation (Months 1-3)

Activities: - Kickoff and project planning with AMC and DNP - Detailed site assessment (tank survey, baseline nitrification testing, existing sensor validation) - Historical data collection and baseline performance analysis - Wastewater characterisation sampling (2-week campaign for BOD/TSS/COD/TN/TP profiles) - Equipment procurement and detailed specifications for edge hardware - **SCADA software architecture design** (development of custom platform replacing Awatech interface) - Model architecture design (all 20 models, algorithms, inter-dependencies) - Data integration planning (Endress+Hauser sensor network API, HART multiplexing) - Operator training curriculum development (IFAS-specific, new software features)

Deliverables: - Project execution plan - Data foundation for model training - Algorithm specifications for all 20 models with inter-dependency documentation - SCADA software architecture and interface mockups - HART sensor integration specifications

Phase 2: Hardware Installation & SCADA Development (Months 4-6)

Activities: - Install edge hardware infrastructure (computing servers, networking, cybersecurity) - **Develop SCADA software platform** (Logicrest-developed system, complete with visualisation, control logic, alarm management) - Integrate SCADA with Endress+Hauser sensor data streams (HART protocol configuration) - Deploy HART protocol multiplexing infrastructure (handle 30+ instruments) - All sensor data now flowing to custom SCADA software (replaces previous Awatech manual interface) - Cybersecurity hardening and protocols implementation - Operator orientation and initial training on new software platform - Site acceptance testing for hardware and software infrastructure

Deliverables: - SCADA software operational with real-time data visualisation - All sensors successfully integrated and data streaming - Hardware infrastructure operational and tested - Operator training materials finalised (software-specific)

Phase 3: Core Model Deployment & Testing (Months 7-9)

Activities: - Deploy Models 1-8 in advisory mode (AI recommendations displayed, operator makes decisions) - Real-time model tuning and refinement using live plant data - Comparative performance analysis (AI recommendations vs. current operations) - Validation against baseline performance - Dashboard and interface refinement based on operator feedback - Begin collecting biofilm performance data (IFAS media establishing)

Deliverables: - Core models (1-8) operational in advisory mode within SCADA - Operator confidence dashboard ready - Baseline performance metrics established - Operator feedback incorporated into interface

Phase 4: Advanced Modules & Optimisation (Months 10-12)

Activities: - Deploy IFAS-specific Models 9-14 in advisory mode - Deploy advanced Models 15-20 (bulking, digester, nutrient, chlorine) - Model calibration with real biofilm performance data and operating patterns - Temperature-dependent nitrification profile establishment (including winter data if available) - Transition to assisted control mode (AI controls specific subsystems: DO, RAS, WAS; operator monitors) - Extended stability testing (minimum 90 days) - Operator confidence building through incremental automation - Performance validation against Year 1 targets

Deliverables: - All 20 models operational and calibrated - Assisted control mode stable and tested - 90+ day continuous operation record - Performance metrics on track for Year 1 targets

Phase 5: Full Autonomy & Stabilisation (Months 13-15)

Activities: - Transition to full autonomous control mode (all models active, minimal operator override) - 24/7 autonomous operation commencement - Final acceptance testing against all performance criteria - Extended performance validation against targets - Continuous model retraining and Optimisation - Documentation of operational procedures - Final operator training and mastery assessment - Project handover and acceptance

Deliverables: - All 20 models operational in autonomous mode - 24/7 operation demonstrated and stable - Performance metrics meet or exceed all Year 1 targets - Complete operational documentation - Project handover complete

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8. EXPECTED ADVANTAGES & BENEFITS

Operational Benefits

Energy Efficiency Improvements

- **Blower energy Optimisation:** 15-30% reduction through adaptive DO control and flow-responsive aeration
- **Specific energy reduction:** Target 0.15-0.20 kWh/m³ (from current ~0.22 kWh/m³)
- **RAS/WAS pump efficiency:** Improved through coupled MLSS control
- **Digester mixing Optimisation:** Reduced over-mixing saves energy while maintaining biogas yield
- **Peak demand reduction:** Time-of-use Optimisation where applicable

Chemical Usage Optimisation

- **Polymer consumption:** 15-25% reduction through intelligent dosing algorithms (vs. fixed overdose)
- **$FeCl_3$ Optimisation:** 15-20% reduction in coagulant with improved flocculation coupling
- **Chlorine Optimisation:** 15-25% reduction (vs. current fixed-dose protocol)
- **Overall cost reduction:** Data-driven approaches eliminate waste and overdosing safety margins

IFAS-Specific Treatment Improvements

- **Total Nitrogen removal:** Reliable <10 mg/L outlet through nitrification-denitrification Optimisation
- **Ammonia-N control:** Consistent <5 mg/L outlet with winter stability (biofilm provides base population)
- **Biofilm health management:** Proactive monitoring prevents degradation and treatment loss
- **Nitrifier stability:** Winter operation maintained at 75-80%+ efficiency (vs. typical 50-60% without seasonal control)
- **Sludge bulking prevention:** Early warning systems enable preventive action
- **Biosolids quality:** Better Stabilisation and pathogen reduction through optimised digestion

Regulatory Compliance & Quality

- **Regulatory Compliance:** 95% Year 1, 99%+ Year 2+ adherence to NGT 2019 standards (vs. current 85-92%)
- **BOD Compliance:** Consistent <5 mg/L outlet (vs. current variable 8-15 mg/L)

- **TN Compliance:** Reliable <15 mg/L outlet (vs. current 15-25 mg/L non-compliant)
- **TSS Compliance:** Consistent <10 mg/L outlet maintenance
- **System Reliability:** Predictive maintenance reduces unexpected failures by 40-60%
- **Biofilm Stability:** Asset protection through real-time health monitoring

Operational Excellence

- **Consistent Performance:** 24/7 autonomous control with minimal skill variability impact
- **Real-time Monitoring:** Comprehensive visibility into plant operations (30+ sensor parameters)
- **Predictive Insights:** Early warning systems for equipment issues and process anomalies
- **Reduced Manual Burden:** Operators transition from reactive crisis management to strategic oversight roles
- **Knowledge Preservation:** AI captures and operationalises senior operator expertise through documented algorithms
- **Process Transparency:** Explainable AI recommendations build operator confidence and enable learning

Technology & Institutional Benefits

- **Technology Leadership:** Positions Ahmedabad as pioneer in IFAS-optimised AI wastewater treatment
- **Scalability:** Modules directly applicable to other IFAS retrofits and new treatment plants nationally
- **Competitive Advantage:** First IFAS-specific AI control system in Indian municipal water treatment
- **Data-Driven Operations:** Establishes foundation for continuous process improvement
- **Institutional Knowledge:** Documented, repeatable, and automated processes ensure continuity

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9. TECHNICAL SPECIFICATIONS SUMMARY

Performance Target

Metric	Current Baseline	Year 1 Target	Year 2+ Target	Measurement
Energy Consumption (kWh/m^3)	0.22	0.18-0.20	0.15-0.18	Power meters on blower, pumps
NGT Compliance Rate	85-92%	95%	99%+	Monthly lab tests (BOD, TN, TSS)
Effluent BOD (mg/L)	8-15 variable	<5 consistent	<5 consistent	CAS80E outlet analyser
Effluent TN (mg/L)	15-25	<15 reliable	<15 reliable	Lab monthly verification
Effluent Ammonia-N (mg/L)	5-10	<5	<5	CAS40D analyser trend
System Uptime	~90%	95%+	98%+	SCADA monitoring
Automation Level	~20% (fixed aeration)	~90%	98%+	% autonomous decisions
Nitrification Winter Efficiency	50-60% (10-15°C)	75-80% (with seasonal SRT)	80%+ (optimised)	Model 9 predictions vs. actual
Biofilm Media Health Status	Unknown/ unmeasure d	Real-time monitoring via Model 11	Predictive lifecycle management	OTE efficiency tracking
Chemical Usage (Polymer kg/day)	Baseline X	X × 0.80-0.85	X × 0.75-0.80	Dosing pump flowmeter

Metric	Current Baseline	Year 1 Target	Year 2+ Target	Measurement
Chlorine Residual Control (mg/L)	1.8-2.2 (over-dosed)	1.2-1.5 (optimised)	1.0-1.3 (efficient)	Chlorine analyser reading

Technical Capabilities

- **Real-time data processing** from 30+ installed process instruments (Endress+Hauser sensor network)
- **Multi-model ensemble prediction** for robust forecasting (20 specialised models with inter-dependency logic)
- **Automated anomaly detection and alerting** with biofilm-specific and process-specific anomaly classification
- **Explainable AI recommendations** for operator acceptance (SHAP value explanations of each decision)
- **Biofilm-aware process control** with distributed oxygen transfer Optimisation (accounting for suspended + biofilm phases)
- **Cybersecurity protocols** (encryption, access control, audit logs, HART protocol security)
- **Hybrid storage** with on-premises backup (data sovereignty, no cloud-only dependency)
- **Seasonal and temperature-adaptive algorithms** (auto-calibration based on historical patterns)
- **Inter-dependent parameter Optimisation** (Model 13 couples MLSS-SRT-F/M; Model 14 couples Temperature-SRT)
- **Custom SCADA software platform** (Logicrest-developed, replaces previous Awatech manual interface)

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9. IMPLEMENTATION ROADMAP

Timeline Overview

```
MONTH 1-3:    Design & Foundation
              ├── Site survey & sensor validation
              ├── Historical data collection
              ├── SCADA architecture design
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```
    └─ Model architecture design (20 modules)

MONTH 4-6:   Hardware & SCADA Integrated Software Development
    ├─ Edge hardware deployment
    ├─ SCADA integrated software development & integration
    ├─ Sensor data flowing to custom platform
    └─ Operator introduction to new software

MONTH 7-9:   Core Model Deployment & Testing
    ├─ Models 1-8 advisory mode
    ├─ Real-time model tuning
    └─ Baseline performance validation

MONTH 10-12:  Advanced Modules & Optimisation
    ├─ Models 9-20 deployment
    ├─ Assisted control transition
    ├─ 90-day stability proof
    └─ Performance validation vs. Year 1 targets

MONTH 13-15:  Full Autonomy & Stabilisation
    ├─ Full autonomous control activation
    ├─ 24/7 operation demonstrated
    ├─ Final acceptance testing
    └─ Project handover
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10. GOVERNANCE & SUPPORT

Governance Structure

- **Steering Committee:** Monthly meetings with AMC, DNP, and Logicrest representatives
- **Technical Working Group:** Bi-weekly technical reviews and issue resolution
- **Documentation:** Regular updates to operational procedures and process documentation

Implementation Support

On-site Support (During Critical Phases): - On-site project manager (50% dedicated time) - Technical support team during installation and integration phases - Remote access for monitoring and troubleshooting - Phase gate reviews before proceeding to next stage

Post-Implementation Support: - 12-month 24/7 help-desk support (model-specific expertise) - Performance monitoring and continuous Optimisation - Quarterly review meetings with stakeholder alignment - Model retraining with new operational data - Seasonal Optimisation (quarterly model updates)

Service Level Agreements: - **System Uptime:** 95%+ availability - **Critical Issue Resolution:** <4 hours for system failures - **Monthly Reports:** Comprehensive analytics and Optimisation recommendations - **Warranty:** 12 months post-handover (parts and labor) - **Extended Support:** Optional annual maintenance contracts available

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11. APPENDICES

Appendix A: Technical Proposal

- 270+ page comprehensive technical documentation
- Complete SCADA software architecture
- All 20 AI/ML model specifications
- Validation protocols and acceptance criteria

Appendix B: IFAS-Specific Technical Documentation

- Biofilm performance control framework
- Temperature-dependent nitrification kinetics
- Winter operation guidelines
- Biofilm stress detection protocols

Appendix C: Case Studies & References

- Similar IFAS installations with AI Optimisation
- Industry benchmarking data
- Published research on IFAS Optimisation

Appendix D: Regulatory Framework

- NGT 2019 Standards and compliance requirements
- GPCB guidelines and monitoring requirements
- State-level pollution control regulations

Appendix E: Team & Expertise

- Logicrest team profiles and credentials
- IITRAM collaboration details

- Advisory board members

Appendix F: Equipment & Vendor Details

- Endress+Hauser sensor specifications (existing instruments)
- SCADA software architecture documentation
- HART protocol specifications
- Integration specifications for edge hardware

Appendix G: Training & Documentation

- Operator training curriculum (22 hours initial + 8 hours annual refresh)
- Standard operating procedures (IFAS-specific)
- User manuals and troubleshooting guides
- System maintenance and Optimisation procedures

CONCLUSION

The Pirana 200 MLD Sewage Treatment Plant, with its recent IFAS upgrade, represents a critical opportunity to transform operational performance through comprehensive AI-driven automation specifically engineered for the plant's enhanced treatment infrastructure.

Strategic Benefits

Operational Transformation: - 20 specialised AI models optimise every treatment process - IFAS-specific modules capture biofilm-based treatment advantages - Real-time biofilm health monitoring protects capital investment - Winter operation optimisation ensures year-round compliance

Environmental & Regulatory Excellence: - 95% → 99%+ NGT compliance progression - Reliable <10 mg/L Total Nitrogen through IFAS nitrification-denitrification - Predictive maintenance reduces environmental incidents - Winter stability prevents seasonal compliance failures

Operational Resilience & Knowledge: - Predictable performance independent of operator skill level - Documented, repeatable, automated processes - Operators transition from crisis management to strategic optimisation - Biofilm-aware controls leverage scientific advantages of IFAS upgrade

Technology & Market Leadership: - Pioneer position in IFAS-optimised AI treatment optimisation - Replicable model for national IFAS retrofit market - Biofilm-specific intelligence differentiates from generic STP optimisation - Continuous improvement foundation enabled

Implementation Excellence

The 15-month phased roadmap delivers: - Robust performance validation before full autonomy - Operator confidence through incremental automation - Real biofilm performance data capture for model refinement - Sustainable operational procedures - Complete SCADA software platform replacing previous manual interfaces

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IMMEDIATE NEXT STEPS

1. **Technical Deep Dive:** Comprehensive review of 20-model architecture and SCADA design
2. **Site Alignment:** Finalise sensor integration, SCADA deployment points, seasonal testing schedule
3. **Stakeholder Alignment:** Joint workshop with AMC and DNP on phased deployment
4. **Contract Negotiation:** Finalise scope and project terms
5. **Project Kickoff:** Commence Phase 1 mobilisation
